Systems/Circuits

Synaptic Changes in Pallidostriatal Circuits Observed in the Parkinsonian Model Triggers Abnormal Beta Synchrony with Accurate Spatio-temporal Properties across the Basal Ganglia

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Excessive oscillatory activity across basal ganglia (BG) nuclei in the β frequencies (12–30 Hz) is a hallmark of Parkinson's disease (PD). While the link between oscillations and symptoms remains debated, exaggerated β oscillations constitute an important biomarker for therapeutic effectiveness in PD. The neuronal mechanisms of β-oscillation generation however remain unknown. Many existing models rely on a central role of the subthalamic nucleus (STN) or cortical inputs to BG. Contrarily, neural recordings and optogenetic manipulations in normal and parkinsonian rats recently highlighted the central role of the external pallidum (GPe) in abnormal β oscillations, while showing that the integrity of STN or motor cortex is not required. Here, we evaluate the mechanisms for the generation of abnormal β oscillations in a BG network model where neuronal and synaptic time constants, connectivity, and firing rate distributions are strongly constrained by experimental data. Guided by a mean-field approach, we show in a spiking neural network that several BG sub-circuits can drive oscillations. Strong recurrent STN-GPe connections or collateral intra-GPe connections drive γ oscillations (>40 Hz), whereas strong pallidostriatal loops drive low-β (10–15 Hz) oscillations. We show that pathophysiological strengthening of striatal and pallidal synapses following dopamine depletion leads to the emergence of synchronized oscillatory activity in the mid-β range with spike-phase relationships between BG neuronal populations in-line with experiments. Furthermore, inhibition of GPe, contrary to STN, abolishes oscillations. Our modeling study uncovers the neural mechanisms underlying PD β oscillations and may thereby guide the future development of therapeutic strategies.

Key words: **basal ganglia; beta oscillation; globus pallidus; pallidostriatal; Parkinson's disease; spiking neural network**

Significance Statement

In Parkinson's disease (PD), neural activity in subcortical nuclei called the basal ganglia (BG) displays abnormal oscillatory synchronization that constitutes an important biomarker for therapeutic effectiveness. The neural mechanisms for the generation of these oscillations remain unknown. Here, in a theoretical neuronal network model strongly constrained by anatomical and physiological data, we show that specific circuit modifications in BG connectivity during PD lead to the emergence of synchronized oscillatory activity in the network with properties that strongly agree with available experimental evidence. This and future theoretical investigations of the neural mechanisms underlying abnormal neuronal activity in PD are necessary to guide the future development of therapeutic strategies to ameliorate symptoms.

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Introduction

Dysfunctions of the basal ganglia (BG)-cortical network are centrally involved in movement disorders such as Parkinson's disease (PD). According to the classical model of BG dysfunction (Albin, 1989; DeLong, 1990), degeneration of dopaminergic neurons in PD induces an imbalance between direct and indirect BG pathways, over-inhibiting the motor system and thereby causing hypo-kinetic motor symptoms. Accordingly, inactivation of internal pallidum (GPi) or subthalamic nucleus (STN) palliates PD symptoms (Bergman et al., 1990; Benazzouz et al., 1993), whereas activating the indirect pathway generates hypokinesia in rodents (Kravitz et al., 2010). This classical model has recently been challenged. First,

the external pallidum (GPe) contains at least two cell-types with different functions (Mallet et al., 2012; Hernández et al., 2015; Fujiyama et al., 2016): the prototypic (forming the indirect pathway) and the arkypallidal (only projecting to striatum) neurons. Second, BG nuclei display strong modifications in firing patterns in PD, with excessive oscillatory synchrony in the β frequency range (12–30 Hz) (Bergman et al., 1994; Bevan et al., 2002; Hutchison et al., 2004). The link between β oscillations and motor symptoms remains a matter of debate; while apparently not causal to motor symptoms (Leblois et al., 2007; Degos et al., 2009; Quiroga-Varela et al., 2013), they are correlated with akinesia/rigidity levels (Kühn et al., 2006; Hammond et al., 2007; Kühn et al., 2009; Sharott et al., 2014; Neumann et al., 2016). Nevertheless, the β power in STN or pallidal local field potentials serves as an important biomarker for deep brain stimulation in PD patients (Little and Brown, 2020; Tinkhauser et al., 2020). Unlike mean firing rate changes, the generation mechanisms of pathological oscillations remain unknown and several competing models have been proposed (reviewed in Pavlides et al., 2015; Rubin, 2017, but also see McCarthy et al., 2011; Corbit et al., 2016; Adam et al., 2022; West et al., 2022). From a theoretical perspective, any neuronal network incorporating negative feedback loops with delays can generate oscillations, with time constants and delays in the loop constraining the oscillations frequency (Ermentrout et al., 2001). The first BG pattern-generator system identified was the reciprocally connected STN-GPe circuit (Plenz and Kital, 1999; Bevan et al., 2002; Terman et al., 2002; Nevado-Holgado et al., 2010). However, many recurrent inhibitory feedback loops in the BG-cortical network could generate PD-related β-synchronization (Pavlides et al., 2015; Rubin, 2017) and alternative circuit generators have been proposed in the cortex (Brittain and Brown, 2014), the striatum (McCarthy et al., 2011), along the hyperdirect pathway (Leblois et al., 2006), and the recently uncovered striato-pallidal circuits (Mallet et al., 2012; Corbit et al., 2016). While these previous studies were at least partially constrained by available anatomical and physiological evidence, the heterogeneity of the biophysical parameters of the underlying model makes the direct comparison of the various generation mechanisms difficult (although see Pavlides et al., 2015). Recently, neural recordings combined with optogenetic manipulations in normal and parkinsonian rats highlighted the central role of the GPe in the expression of β oscillations, while showing that it does not require the integrity of STN or motor cortex (De la Crompe et al., 2020), ruling out several of the previously suggested mechanisms. This study also highlighted that the phase relationship between various BG neural populations is a critical hint to probe which circuit is generating the β oscillations. Here, we evaluate the mechanisms for generation of abnormal β oscillations in a BG network model where time constants, synaptic strengths, connectivity patterns, and firing rate distributions are strongly constrained by available anatomical and physiological (in vivo and in vitro) rodent data. Guided by a mean-field approach, we show in a spiking neural network (SNN) that the frequency of the pathological β-oscillations and their phase-locking properties across BG nuclei are most likely driven by the interaction between the slow-oscillating pallidostriatal loop and the fast inhibitory loop made by recurrent inhibitory collaterals between GPe prototypical neurons.

Materials and Methods

Model architecture

In this article, we focus on the BG circuits that may generate synchronized oscillatory activity. As highlighted in the introduction, oscillatory activity emerges from inhibitory feedback with time delay. As the output nuclei of the BG (GPi/SNr) do not send significant projections back to other BG nuclei, they do not participate in recurrent feedback circuits. Similarly, D1-expressing medium spiny neurons (D1-MSNs) in the striatum mainly participate in purely feedforward direct pathways and are unlikely to be included in a significant recurrent circuit inside the BG (but see McCarthy et al., 2011). In addition, their activity is strongly reduced after dopamine depletion (DD) (Mallet et al., 2006; Parker et al., 2018). For all these reasons, we do not include output nuclei and D1-MSNs in our model. We include all other major populations of the BG network. In particular, our model reflects on the recently discovered cellular dichotomy between the principal GPe cell population, the prototypic neurons (projecting to STN and (GPi/SNRr)), and the arkypallidal neurons (only projecting to the striatum; Mallet et al., 2012). Our model consists of five subpopulations, four inhibitory and one excitatory: fast-spiking interneurons (FSIs) and D2-MSNs in the striatum, prototypical (Proto) and arkypallidal (Arky) neurons in the GPe, and lastly, STN excitatory neurons (Fig. 1). Henceforth, when referring to these neuronal populations, we simply use Proto, Arky, STN, FSI, and D2.

We include all major connections that connect the considered neuronal populations. They contribute to several closed negative feedback loops and have the Proto subpopulation as their common node. There are four such negative feedback loops. The STN-Proto recurrent network is one which involves Proto and STN, and comprises two synaptic connections, STN to Proto (Kita and Kitai, 1991; Magill et al., 2000) and Proto to STN (Kita et al., 1983; Kita and Kita, 1994; Baufreton et al., 2009) (referred to as STN loop in the figures). Second, there is lateral inhibition within the Proto subpopulation (Sims et al., 2008; Ketzef and Silberberg, 2020) (referred to as the Proto loop). The two remaining are pallidostriatal loops that both include the projection from D2-MSNs to Proto neurons (Kita, 1994; Cooper and Stanford, 2000; Ketzef and Silberberg, 2020; Aristieta et al., 2021). The FSI loop has the additional connections of Proto to FSI (Bevan, 1998; Mallet et al., 2012; Corbit et al., 2016; Saunders et al., 2016) as well as FSI to D2 (Mallet et al., 2005). Lastly, the Arky loop is the one that connects Proto to Arky (Ketzef and Silberberg, 2020; Aristieta et al., 2021), Arky to D2 (Mallet et al., 2012; Fujiyama et al., 2016), and as previously mentioned, D2 to Proto. We have deliberately ignored connections that have been shown to be significantly weaker (or not well characterized), namely the connection from STN to Arky (weaker than STN to Proto; see Ketzef and Silberberg, 2020; Aristieta et al., 2021), from Arky to FSI (as this projection is not well characterized and likely weaker than the selective and potent inhibition from Proto to FSI; Bevan, 1998; Saunders et al., 2016), and the projection from D2 to Arky (weaker than D2 to Proto; see Ketzef and Silberberg, 2020). Indeed, as our aim is to provide a mechanistic understanding of the beta generation circuits, we focused on a simplified network that includes all major BG connections that contribute to creating a closed negative feedback loop within the BG that can give rise to oscillations. We nevertheless simulate a more complete network with all known connections present to check that the dynamics described here remain valid in the presence of these additional pathways (Extended Data Fig. 8-4). As explained below, all parameters of the model are extracted or strongly constrained by experimental data collected in rodents. When available, we use experimental data from rats, otherwise, we use data from mice instead and clearly indicate it in the corresponding tables.

Population size and connectivity

The number of neurons in all populations of the model is scaled down compared to the real neuron population size to make the simulation computationally tractable. We consider a constant number of neurons N^{sim} when simulating all α sub-populations ($N^{\text{sim}} = 1{,}000$ in the spiking model and $N^{\text{sim}} = 100$ in the rate model, see below) and follow the approach proposed by Golomb and Hansel (2000) to set the number of connections population size accordingly. As such, the number $K_{\beta\alpha}^{\text{sim}}$ of synapses received by one neuron in population β from all neurons in population α is determined by

$$\frac{1}{K_{\beta\alpha}} - \frac{1}{N_\alpha} = \frac{1}{K_{\beta\alpha}^{\text{sim}}} - \frac{1}{N^{\text{sim}}}, \tag{1}$$

Table 1. Number of presynaptic neurons converging onto a postsynaptic neuron

Parameter	Value	Reference
$K_{Proto-D2}$	3,100	Kawaguchi et al. (1990)
$K_{STN-Proto}$	442	Baufreton et al. (2009)
$K_{Proto-STN}$	63	Koshimizu et al. (2013)
$K_{Proto-Proto}$	104	Sadek et al. (2005)
$K_{Arky-Proto}$	104	Sadek et al. (2005)
K_{D2-FSI}	51	Guzmán et al. (2003) and Gittis et al. (2011)
$K_{FSI-Proto}$	67	Bevan (1998)
$K_{D2-Arky}$	10	Estimate (see Extended Data 1)

$K_{\beta\alpha}$ represents the number of presynaptic neurons in population α converging onto one postsynaptic neuron in β for all connections in the spiking neuronal BG network. The references on which the estimates are based are reported in the last column. See Extended Data 1 and Table 1-1 for more details on the derivation of $K_{\beta\alpha}$ from experimental observation.

where $K_{\beta\alpha}$ is the number of connections from population α to a neuron in population β and N_α is the size of population α found in the experimental literature (see Extended Data Table 1-1 (Kincaid et al., 1998, Oorschot, 1998, Koós and Tepper, 1999, Tepper et al., 2010)). This scaling ensures that the dynamical properties of the original circuit are preserved in the scaled version. The values of $K_{\beta\alpha}$ derived for each population for the spiking neuronal network with $N^{sim} = 1{,}000$ are reported in Table 1. $K_{\beta\alpha}$ are calculated similarly for $N^{sim} = 100$ in the rate model (values not shown here). Note that henceforth we use the notation β-α to refer to the synaptic projection from population α to population β.

Neuronal model

While most results shown here are computed in a spiking neuronal network model, we will also consider a simplified version of the model in which neurons are represented by their firing rate (Wilson and Cowan, 1972). This model has the advantage that several aspects of its dynamics can be investigated analytically. The knowledge of the properties of this reduced model guides us in the investigation of the detailed spiking model.

Rate model

In order to simulate the firing activity of each neuron, we use a model similar to Leblois et al. (2006), where the dynamics of the activity of a single neuron is characterized by the rate model (Wilson and Cowan, 1972). The spontaneous firing rate of a neuron is given by $A_i^\alpha = F(I_i^\alpha)$, where α denotes the population and can be any of Proto, STN, Arky, D2, or FSI. I_i^α is the total input to neuron i where $i = 1, \ldots, N_{sim}$ and F is its non-linear input–output transfer function. Here for simplicity, the transfer function is considered to be a piecewise linear function $F = [I_i - \theta]_+$, where $[x]_+ = x$ for $x > 0$ and 0 otherwise, with the activity threshold $\theta = 0.1$. Each neuron i in population α has a synaptic contribution of $m_i^{\beta\alpha}$ to each neuron it is connected to in the postsynaptic population β. $m_i^{\beta\alpha}$ is a low-pass filtered version of the instantaneous level of activity A_i^α (Shriki et al., 2003), and its dynamics are governed by

$$\tau^{\beta\alpha} \frac{dm_i^{\beta\alpha}}{dt} = -m_i^{\beta\alpha} + A_i^\alpha, \tag{2}$$

where $\tau^{\beta\alpha}$ is the synaptic decay time constant of the projection from population α to population β. The simulation of the dynamics is integrated using a standard first-order forward Euler method with a time step $dt = 0.1$ ms to ensure maximum precision. Each neuron receives multiple synaptic inputs from the neurons in their presynaptic populations determined by the connection matrix $J^{\beta\alpha}$ *which has binary elements, with one indicating the presence of a connection, and zero the absence of one.* Therefore, we need to sum the inputs from all presynaptic neurons to get the net synaptic input to each postsynaptic neuron:

$$I_i^\beta(t) = \sum_\alpha \sum_{j=1}^{N_\alpha} G_{\beta\alpha} J_{ij}^{\beta\alpha} m_{ij}^{\beta\alpha}(t - \Delta^{\beta\alpha}), \tag{3}$$

where $\Delta^{\beta\alpha}$ is the axonal transmission delay of the projection from population α to population β (see first column in Table 5). $G_{\beta\alpha}$ is the synaptic weight which is considered to be the same for all connections and is the minimum value for which stable oscillations appear (see below for the definition of stable oscillation). We simulate each population with size $N = 100$. We have also performed the simulations with size $N = 1{,}000$ to ensure that the dynamics remain unchanged in our smaller network. Moreover, we have investigated this reduced model analytically in the case of all-to-all connectivity in the network. In this case, the steady states of the network in response to constant inputs can be obtained by solving the fixed-point equations for the dynamics (see "Theoretical analysis"). In these states, the activities are determined by a set of linear equations that can be solved straightforwardly. The conditions for the stability of steady states can then be derived. Indeed, a fixed-point solution of the dynamical equations is stable if any small perturbation around it eventually decays over a large time. If certain perturbations increase with time, the fixed point is unstable. To investigate the stability of a steady state, we study the equations of the dynamics linearized around that state (Strogatz, 2018) (see derivation in "Theoretical analysis"). In particular, the derivation leads to the estimation of the frequencies of the oscillatory instabilities for individual negative feedback loops in the BG network (see "Theoretical analysis" for more detail).

Spiking neural network

The neurons in the SNN are simulated using the leaky integrate and fire (LIF) model (Lapicque, 2007). In this model, the dynamics of the membrane potential of each neuron are given by

$$\tau_m \frac{dV(t)}{dt} = -(V(t) - V_{rest}) + I_{syn}(t) + I_{ext}(t), \tag{4}$$

where $V(t)$ is the membrane potential of each neuron at time t, V_{rest} is the resting membrane potential (RMP), τ_m is the membrane time constant, $I_{syn}(t)$ is the total synaptic input, and $I_{ext}(t)$ is the external input of each individual neuron. Once the membrane potential of the neuron reaches the action potential threshold (APth), a spike is discharged and $V(t)$ is manually reset to the RMP (see "Neuronal model" for more detail on the modified V_{rest} value). The values for the RMP, APth, and τ_m are drawn from Gaussian distributions with respective parameters extracted from the experimental literature and summarized in Tables 2 and 3. We modify the standard deviation (SD) of some of the distributions to fit the

Table 2. Electrophysiological properties of the action potentials

Neuron	RMP (mV)	RMP range	APth (mV)	Reference	σ_{RMP}, σ_{APth} in the model (mV)
D2-MSN	−76.8 ± 4.4	(−100,−55)	−50.0 ± 0.6	Slaght et al. (2004)	3, 0.6
STN	−59.0 ± 1.2	(−75,−55)	−50.8 ± 1.5	Paz et al. (2005)	0.5, 0.5
Arky	−70*	(−90,−60)	−55.0 ± 7.6	Abdi et al. (2015)	1, 2
Proto	−65*	(−85,−60)	−54.8 ± 7.1	Abdi et al. (2015)	1, 1
FSI	−78.2 ± 6.2	(−85,−60)	−52.4 ± 4.6	Schulz et al. (2011)	0.5, 0.5

Columns 2 and 4 show APth and RMP values reported in the references as mean ± SD. In the model, RMP ranges are set arbitrarily while respecting the means. Some SDs are modified to fit the firing rate distribution, and corresponding values used in the model are reported in the last column. The values indicated by the asterisks are estimated from Figure 7 in Abdi et al. (2015).

Table 3. Membrane time constants used in the model

Neuron	τ_m (ms)	Reference	σ_{τ_m} and range in the model (ms)
D2-MSN	4.9 ± 1.6	Slaght et al. (2004)	0.5—(2, 12)
STN	5.1 ± 2.4	Paz et al. (2005)	0.6—(2, 10)
Arky	19.9 ± 13.0	Cooper and Stanford (2000)	3—(2, 100)
Proto	12.9 ± 10.2	Karube et al. (2019)	1.3—(2.25, 45)
FSI	3.1 ± 1.1	Schulz et al. (2011)	0.3—(1, 6)

Second column indicates the membrane time constants as reported in the references (mean ± SD). SDs are modified in the model to fit the firing rate distribution and the corresponding values used in the model are reported in the last column. The values are chosen among different studies reported in Extended Data Table 3-1.

firing rate distributions reported in the experimental recordings as reported in the corresponding tables. There are a few studies that have measured the membrane time constants of the different populations as shown in Extended Data Table 3-1 (Bugaysen et al., 2010, Planert et al., 2013, Russo et al., 2013, Akopian et al., 2016). To choose among those, we have prioritized measurements in rats over mice, in vivo over in vitro, temperatures closer to 37 °C, larger number of recorded cells, more recent studies, as well as measurements in adult animals over juveniles. The Gaussian distribution for the RMP and τ_m are truncated to avoid problems with the numerical integration.

Synaptic inputs are modeled by a double exponential with independent rise and decay time constants (Fourcaud and Brunel, 2002). The sum of synaptic inputs from all presynaptic neurons $I_{\text{syn}}(t)$ to each neuron is given by

$$\tau_{r_j}^{\beta\alpha}\frac{\mathrm{d}I_{r_j}^{\beta\alpha}(t)}{\mathrm{d}t} = -I_{r_j}^{\beta\alpha}(t) + \sum_{i=1}^{N_{sim}} J_{ij}^{\beta\alpha}\sum_{\text{spike}}\delta(t-(t_i^{\text{spike}}-\Delta_j^{\beta\alpha}))\tau_{m_j}^{\beta}, \quad (5)$$

$$\tau_{d_j}^{\beta\alpha}\frac{\mathrm{d}I_{\text{syn}_j}^{\beta\alpha}(t)}{\mathrm{d}t} = -I_{\text{syn}_j}^{\beta\alpha}(t) + I_{r_j}^{\beta\alpha}(t), \quad (6)$$

where $\tau_{r_j}^{\beta\alpha}$ and $\tau_{d_j}^{\beta\alpha}$ are the rise and decay time constants of the synaptic input to neuron j in postsynaptic population β. $\Delta_j^{\beta\alpha}$ is the synaptic transmission delay from populations α to neuron j in population β. $J_{ij}^{\beta\alpha}$ is the connectivity matrix between populations α and β where each nonzero element is equal to the synaptic weight of the corresponding synapse. $I_{r_j}^{\beta\alpha}$ is the external input to neuron j in postsynaptic population β. As indicated by the superscripts, all of the aforementioned values are specific to each projection β–α. For the sake of simplicity, the time constants used to describe the incoming projections from presynaptic population *alpha* converging to neuron j in population β are all identical. However, the values of $\tau_{r_j}^{\beta\alpha}$, $\tau_{d_j}^{\beta\alpha}$, and $\Delta_j^{\beta\alpha}$ are drawn randomly from a Gaussian distribution for each postsynaptic neuron $j = 1, \ldots, N_{sim}$, with means and SD of the underlying distributions extracted from the experimental literature as reported in Table 5. The connection weights $J_{ij}^{\beta\alpha}$ are either zero (with a probability $(N^{\text{sim}} - K_{\beta\alpha}^{\text{sim}})/N^{\text{sim}}$) or drawn from a log-normal distribution with a SD that spans one order of magnitude (Buzsáki and Mizuseki, 2014). The means of the synaptic weight distributions for the STN-Proto, Proto-STN, and Arky-Proto are set such that the network activity resembles those of the β-induction experiments (see "Results" and Fig. 7 for more detail). Due to the lack of such experimental data for the rest of the projections, we set their weights in a way to mimic the physiological healthy states with asynchronous activity across the BG neuronal populations. The means of the synaptic weight distributions for each projection in the healthy state are reported in Table 4.

As only a subset of the synaptic connections to each population is considered here, the external input to each population is tuned to set the firing rate of the neurons to those reported in the experiments in each state (healthy and dopamine-depleted). This external input provided to each neuron is drawn from a Gaussian distribution separately for each state. In order to estimate the mean of this Gaussian distribution, the mean firing rate is computed for a range of external inputs with varying means while holding the SD constant, in turn yielding an empirical I–F curve (input-firing rate). Then, based on whether the desired firing rate is low (<15 spk/s, where the I–F curve is non-linear) or high (>15 spk/s, linear section of the curve), either a sigmoid function or a line is fitted to the simulated data points. Subsequently, the mean input required to obtain the firing rate reported in the experiments for each population is extrapolated from its respective fitted curve. The SD of the external input is then separately tuned to replicate the variability of the firing rate reported in the experimental literature (Table 6). This procedure is separately performed for the healthy and dopamine-depleted states to estimate their corresponding external inputs. Wherever we were unable to replicate the firing rate distribution solely by manipulating the SD of the external input, we tried tuning it by changing the SDs of τ_m and/or RMP and/or APth. The SDs of these parameters are reported in Tables 2 and 3. Concerning the MSNs (D2), the level of activity considered in our model reflects the activated anesthetized state (as in De la Crompe et al., 2020) which is similar to the activity in the awake state. The LIF neuron model cannot account for the bistability of the membrane potential that has been reported in MSNs in vitro as well as in anesthetized animals. Thus, our results are confined to awake or anesthetized active states, in which β oscillations are expressed and MSN bistability has been shown to vanish (Mahon et al., 2001).

Table 4. Mean of the synaptic weight distributions

Synaptic weight	Mean (mV)
$G_{\text{Proto}-\text{D2}}$	−0.279
$G_{\text{STN}-\text{Proto}}$	−0.096*
$G_{\text{Proto}-\text{STN}}$	0.061
$G_{\text{Proto}-\text{Proto}}$	−0.032*
$G_{\text{Arky}-\text{Proto}}$	−0.112
$G_{\text{D2}-\text{FSI}}$	−0.737
$G_{\text{FSI}-\text{Proto}}$	−0.709
$G_{\text{D2}-\text{Arky}}$	−0.268

Values correspond to the mean of the log-normal distribution for the synaptic weight of each projection β–α in the healthy state simulated in the spiking network. Values marked by asterisks are doubled in the dopamine-depleted state.

Membrane potential initialization. Spurious synchrony can be present in the network due to the initialization of the membrane potentials of neurons (setting $V(t=0)$). Such spurious synchrony would be mirrored in other synaptically connected populations and interfere with the network oscillations which we are interested to look at. We, therefore, run a 2-s long simulation to let the network reach its stable dynamical state. We then compute the distribution of membrane potential values of all the neurons for the entirety of that simulation and initialize all subsequent simulations by randomly drawing from the corresponding membrane potential distributions.

Numerical integration. We use the second-order Runge–Kutta (RK2) method to numerically integrate the membrane potential at each time step. However, the discontinuity of the LIF model at the spike times can cause numerical errors. So the time steps over which the equation is integrated should be kept small in order to keep the errors to a minimum. This can slow down the simulation dramatically. As a solution, we use a linear interpolation method proposed by Hansel et al. (1998) when updating the membrane potential at spike times. The interpolated value of the membrane potential at the time step after the spike time, $V(t+\Delta t)$, is given by

$$V(t+\Delta t) = (V_0(t+\Delta t) - V_{\text{th}})\left(1 + \frac{\Delta t}{\tau_m}\frac{V(t) - V_{\text{rest}}}{V_0(t+\Delta t) - V(t)}\right) + V_{\text{rest}}, \quad (7)$$

where $V_0(t+\Delta t)$ is calculated by the standard RK2 at the spike time and V_{th} is the APth. In this RK2 method with linear interpolations, the potential is reset to the value given by Equation 7 instead of V_{rest}. When using the RK2 integration with the aforementioned interpolation, a time step of 0.25 ms results in a 10% error in coherence as compared to the exact analytical solution, whereas 0.063 results in a 1% error (Hansel et al., 1998). With this information, we chose to use a time step of 0.1 ms in our simulations. We also performed sample simulations with a 0.01 ms time step. Since the oscillations are not stable, there are some deviations between the average firing rate traces of the full network simulations. Nevertheless, we did not observe any significant change in the resulting dynamics (Extended Data Fig. 5-1).

Noise. We model the synaptic noise of each neuron as an Ornstein–Uhlenbeck process to account for the temporal dynamics of the noise element that decays with time. At each time step, the value of noise is computed by

$$\eta(t+\Delta t)=\left[\frac{-\eta(t)}{\tau_{\text{noise}}}+\sqrt{\frac{2}{\tau_{\text{noise}}\Delta t}}\xi(t)\right]\Delta t, \tag{8}$$

where $\tau_{\text{noise}}=10$ ms and $\xi(t)$ is sampled from a normal distribution $\xi(t)\sim\mathcal{N}(\mu=0,\ \sigma^2_{\text{noise}})$independently at every time step.

Data analysis

Population firing rate

The firing rate of all the neurons in each population is averaged in each time bin to yield the average population firing rate as shown in the figures. In the SNN, the average population firing rate is further smoothed using a moving average window of 5 ms.

Power spectrum

The power spectrum of the average firing rate activities is derived using the Welch method implemented in the *scipy.signal* library (Welch, 1967) using 50%-overlapping, 1-second long windows. The simulations are run for 20.3 and 25.3 second in the case of the rate model and the SNN, respectively. The first 300 ms is discarded to prevent the contribution of the transient dynamics that depend on initialization. In addition, the SNN simulations are repeated three times, each with a different network initialization to yield a mean and standard error of the mean (SEM) for each frequency bin when measuring the power spectrum.

Phase histogram

The β-filtered (12-30 Hz) mean population firing rate of the Proto neurons is used as the reference signal to obtain the linear phase histograms. The phase of each spike is measured within the cycle created by the two closest peaks of the reference signal in time, one before and the other after. Then, the linear phase histogram of the spikes from each neuron is calculated. Finally, the phase histogram of all neurons is subjected to Rayleigh's test and only those that pass are taken into account. The value at which the phase histogram is maximized for each neuron is used to create the box plots in the phase histogram plots.

Cross-correlation

The spike cross-correlations (CCs) is calculated using the *scipy.signal.correlate* function in Python using the "full" mode. Two thousand random pairs of neurons are chosen for each β–α connection without pair replacement. Then, the CC for each pair is individually calculated using the 25-s long spike trains with 0.1 ms bin size time resolution. Subsequently, the results of cross-correlation for all the pairs are reported as mean ± SEM. The CCs were sampled for every 2.5 ms of lag and plotted across the $[-0.2,+0.2]$ second lag window in order to compress the data presented in the plots.

Coherence

The coherence is measured using the *scipy.signal.coherence* function in Python. One thousand random pairs of neurons are chosen for each β–α connection without pair replacement. Then, the coherence for each pair is individually calculated using the 25-s long spike trains with 0.1 ms bin size time resolution. The results of coherence for all the pairs are reported as mean ± SEM for every frequency bin.

Stable oscillation

We introduce a metric to help us detect the transition boundary between the steady state and the stable oscillatory regime in the rate model. This metric is defined as the ratio of the peak amplitudes of the firing activity in the last cycle relative to the second cycle. Once this metric reaches the value of 1.01 and is still lower than 1.2, it is considered that the network has transitioned to a stable oscillatory regime.

Theoretical analysis. In this section, we aim to analytically derive the frequencies of the oscillatory instabilities for individual isolated loops of the model network. For the sake of simplicity, we consider an all-to-all connectivity matrix $J^{\beta\alpha}=I$. In this case, the steady states of the network in response to constant inputs can be obtained by solving the fixed point of Equation 2. For steady states where all neuronal populations have inputs above the threshold, the activities are determined by a set of linear equations that can be solved straightforwardly. The conditions for the stability of steady states can then be derived. Indeed, a fixed-point solution of the dynamical equations is stable if any small perturbation around it eventually decays at large time. If certain perturbations increase with time, the fixed point is unstable. To investigate the stability of a steady state, we study the equations of the dynamics linearized around that state (Strogatz, 2018). We consider a perturbation term for the output $m^{\beta\alpha}$ described in Equation 2.

$$m^{\beta\alpha}=m^{\beta\alpha}_{\text{steady}}+\delta^{\beta\alpha}e^{\lambda t}, \tag{9}$$

replacing this in Equation 2 we get

$$\tau^{\beta\alpha}\lambda\delta^{\beta\alpha}e^{\lambda t}=-(m^{\beta\alpha}_{\text{steady}}+\delta^{\beta\alpha}e^{\lambda t})+A^{\alpha}. \tag{10}$$

Then, for an activation function with unit gain, we have $A^{\alpha}=I^{\alpha}$. Therefore, if we consider a network of only two populations α and β (Fig. 2*B*), from Equation 2 we have

$$A^{\alpha}=G_{\alpha\beta}(m^{\alpha\beta}_{\text{steady}}+\delta^{\alpha\beta}e^{\lambda(t-\Delta^{\alpha\beta})}). \tag{11}$$

At steady state, from Equation 2 we can write $m^{\beta\alpha}_{\text{steady}}=G_{\beta\alpha}m^{\alpha\beta}_{\text{steady}}$. Therefore, replacing A^{α} in Equation 10 we have

$$e^{\lambda t}(1+\lambda\tau^{\beta\alpha})\delta^{\beta\alpha}-e^{\lambda t}(e^{-\lambda\Delta^{\alpha\beta}}G_{\alpha\beta})\delta^{\alpha\beta}=0, \tag{12}$$

$$\rightarrow\begin{pmatrix}-e^{-\lambda\Delta^{\alpha\beta}}G_{\alpha\beta} & 1+\lambda\tau^{\beta\alpha}\\ 1+\lambda\tau^{\alpha\beta} & -e^{-\lambda\Delta^{\beta\alpha}}G_{\beta\alpha}\end{pmatrix}\begin{pmatrix}\delta^{\alpha\beta}\\ \delta^{\beta\alpha}\end{pmatrix}=0, \tag{13}$$

$$P(\lambda)=Det(M)=e^{-\lambda(\Delta^{\beta\alpha}+\Delta^{\alpha\beta})}G_{\beta\alpha}G_{\alpha\beta}-(1+\lambda\tau^{\beta\alpha})(1+\lambda\tau^{\alpha\beta})=0. \tag{14}$$

The solutions to this equation are in general complex numbers. The steady state is stable provided that $\text{Re}(\lambda)<0$ for all the solutions. It is unstable if at least one solution with $\text{Re}(\lambda)>0$ exists. If, for this solution, $\text{Im}(\lambda)<0$, the system undergoes a non-oscillatory instability. If $\text{Im}(\lambda)>0$, the instability is a Hopf bifurcation (Strogatz, 2018) at a frequency $\text{Im}(\lambda)$. For such an oscillatory instability $\lambda=i\nu$, we get

$$\begin{aligned}P(i\nu)=(G^{*}_{2N}\cos\Delta^{+}_{2N}\nu+\tau^{*}_{2N}\nu^2-1)-i(G^{*}_{2N}\sin\Delta^{+}_{2N}\nu+\nu\tau^{+})=0\\ \rightarrow G^{*}_{2N}=\tfrac{-\nu\tau^{+}_{2N}}{\sin\Delta^{+}_{2N}\nu},\\ \rightarrow\tau^{*}_{2N}\nu^2-\tau^{+}_{2N}\nu\cot\Delta^{+}_{2N}\nu-1=0\end{aligned} \tag{15}$$

where

$$\begin{aligned}G^{*}_{2N}&=G_{\beta\alpha}G_{\alpha\beta}\\ \tau^{*}_{2N}&=\tau^{\beta\alpha}\tau^{\alpha\beta}\\ \tau^{+}_{2N}&=\tau^{\beta\alpha}+\tau^{\alpha\beta}\\ \Delta^{+}_{2N}&=\Delta^{\beta\alpha}+\Delta^{\alpha\beta}.\end{aligned} \tag{16}$$

In particular, we can derive the resultant frequency for a feedback loop with three nodes (Fig. 2C) (i.e., Proto-FSI/Arky-D2) as follows:

$$\begin{pmatrix} 1+\lambda\tau_{\beta\alpha} & 0 & -e^{-\lambda\Delta^{\alpha\gamma}}G^{\alpha\gamma} \\ -e^{-\lambda\Delta^{\beta\alpha}}G^{\beta\alpha} & 1+\lambda\tau_{\gamma\beta} & 0 \\ 0 & -e^{-\lambda\Delta^{\gamma\beta}}G^{\gamma\beta} & 1+\lambda\tau_{\alpha\gamma} \end{pmatrix} \begin{pmatrix} \delta^{\beta\alpha} \\ \delta^{\gamma\beta} \\ \delta^{\alpha\gamma} \end{pmatrix} = 0 \quad (17)$$

$$\begin{aligned} P(\lambda) &= 1+\lambda\tau_{3N}^{+}+\lambda^2\tau_{3N}^{*+}+\lambda^3\tau_{3N}^{*}-G^{*}e^{-\lambda\Delta^{+}}=0 \\ &\rightarrow G^{*}=\frac{1-\nu^2\tau_{3N}^{*+}}{\cos\nu\Delta_{3N}^{+}} \\ &\rightarrow \tau_{3N}^{*}\nu^3+(\tau_{3N}^{*+}\nu^2-1)\tan\nu\Delta_{3N}^{+}-\tau_{3N}^{+}\nu=0 \end{aligned} \quad (18)$$

where

$$\begin{aligned} G_{3N}^{*} &= G_{\beta\alpha}G_{\alpha\gamma}G_{\gamma\beta} \\ \tau_{3N}^{*} &= \tau^{\beta\alpha}\tau^{\alpha\gamma}\tau^{\gamma\beta} \\ \tau_{3N}^{+} &= \tau^{\beta\alpha}+\tau^{\alpha\gamma}+\tau^{\gamma\beta} \\ \tau_{3N}^{*+} &= \tau^{\gamma\beta}\tau^{\alpha\gamma}+\tau^{\alpha\gamma}\tau^{\beta\alpha}+\tau^{\beta\alpha}\tau^{\gamma\beta} \\ \Delta_{3N}^{+} &= \Delta^{\beta\alpha}+\Delta^{\alpha\gamma}+\Delta^{\gamma\beta}. \end{aligned} \quad (19)$$

We can also calculate the frequency of oscillation for a neuron that contacts itself (Fig. 2A) to mimic the Proto lateral inhibition

$$\begin{aligned} P(\lambda) &= e^{-\lambda\Delta^{\alpha\alpha}}G_{\alpha\alpha}-\lambda\tau^{\alpha\alpha}-1=0 \\ &\rightarrow \tau^{\alpha\alpha}\nu+\tan\nu\Delta^{\alpha\alpha}=0. \end{aligned} \quad (20)$$

In order to estimate the oscillation frequencies of each of the individual loops, we have solved the corresponding characteristic equations using *scipy.optimize.fsolve* function in Python (Fig. 3A).

Code availability

The simulations of the network model and the data analysis are all performed using custom code in Python available in this Github repository.

Results

Building on the abundant anatomical and physiological data from rodents in normal and parkinsonian conditions and the well-established abnormal β-band synchronization present in the BG of parkinsonian rats, we study how various negative feedback loops inside a model network of rat BG circuits may generate abnormal β-band oscillations. We follow a two-step approach (see "Materials and Methods" for the details). We first rely on a simplified model where neuronal activity is represented by its firing rate and a limited number of parameters determine the frequency of oscillations in each closed-loop circuit. Guided by the analysis of this reduced model, we then analyze the properties of a spiking neuronal network model with the same architecture where each neuron is represented by its membrane potential. This realistic model allows us to faithfully represent the firing properties of BG neurons and is a more accurate representation of the circuit dynamics underlying normal and pathological BG activity. Recent evidence shows that the GPe is a necessary hub for the generation of the β oscillations in parkinsonian rats (De la Crompe et al., 2020), and our model, therefore, includes all closed-loop BG circuits that include at least one GPe neuronal population. The GPe consists of two distinct subpopulations, the prototypical (Proto) and the arkypallidal (Arky) neurons (Mallet et al., 2012). The prototypical neurons send projections to the striatum, contacting mainly the FSIs (Bevan, 1998; Mallet et al., 2012; Corbit et al., 2016; Saunders et al., 2016). This connection together with projections from FSIs to D2-MSNs and D2-MSNs back to Proto neurons creates a candidate feedback loop (Corbit et al., 2016), the FSI loop. Similarly,

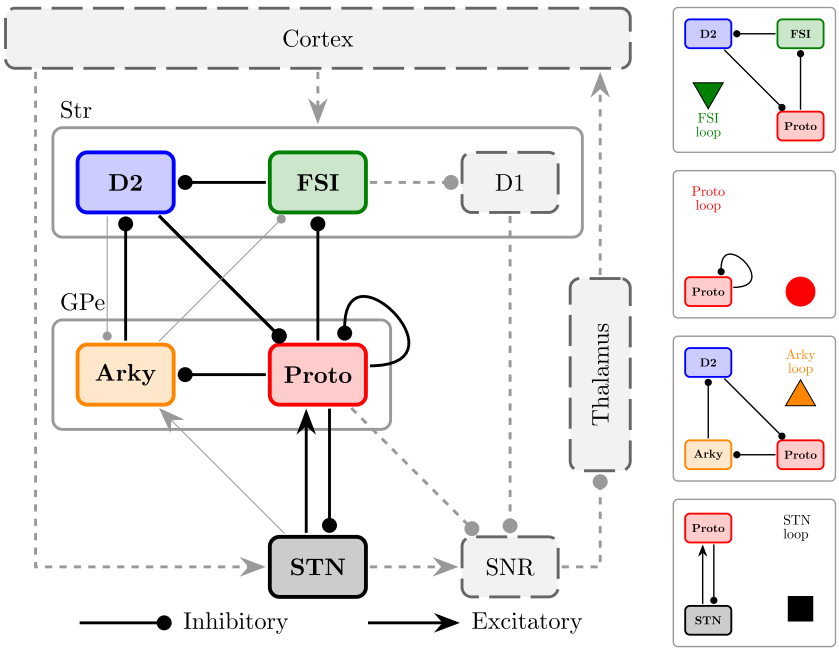


Figure 1. The model architecture. The model includes prototypical (red) and arkypallidal (orange) populations of the external globus pallidus (GPe), D2-expressing medium spiny neurons (D2-MSNs, blue), and fast-spiking interneurons (FSIs, green) within the striatum, and finally the STN (black). Arrows/dots indicate excitatory/inhibitory connections, respectively. Black connections among the colored populations make up four different closed negative feedback loops represented on the right: FSI loop (green triangle), Proto loop (red circle), Arky loop (orange triangle), and STN loop (black square). Light gray regions and connections are not included in the main model; however, the thin solid gray connections (Arky-D2/STN, FSI-Arky) have been included in a supplementary simulation (Extended Data Fig. 8-4).

Arky neurons send projections to the striatum and contact D2-MSNs (Mallet et al., 2012; Fujiyama et al., 2016). Therefore, considering the lateral inhibition from Proto to Arky creates another pallidostriatal loop, the Arky loop. The well-established recurrent network between the STN and GPe (Proto) (Plenz and Kital, 1999; Terman et al., 2002; Nevado-Holgado et al., 2010; Kumar et al., 2011; Pasillas-Lépine, 2013; Nevado-Holgado et al., 2014; Pavlides et al., 2015; Shouno et al., 2017; Koelman and Lowery, 2019; Chen et al., 2020) and the recurrent lateral inhibition between Proto neurons (Ketzef and Silberberg, 2020; Aristieta et al., 2021) also constitute negative feedback loop circuits. All the loops and connections are represented in Figure 1.

Wide range of frequencies from BG oscillation-generator circuits

As mentioned in the introduction, a neuronal circuit forming a closed negative feedback loop can generate oscillations if the strength of its feedback is sufficient (Ermentrout et al., 2001). There are four different negative feedback loops in our BG network model that can generate oscillations, as detailed above (FSI-D2-Proto loop, D2-Proto-Arky loop, STN-Proto loop, and Proto recurrent loop; see Fig. 1). Each loop can be considered as a dynamical system with a steady state (or fixed point) in which all populations have a constant activity level, which is stable if any small perturbation around it eventually decays over time, unstable otherwise. The stability of a steady state can be studied by solving the equations of the dynamics linearized around that state (Strogatz, 2018). In the case of negative feedback loops, the stable fixed-point loses stability via a Hopf bifurcation for sufficiently large overall gain, defined as the product of the synaptic weights along the connections of the loop (see "Materials and Methods, Theoretical analysis" for the detailed calculation). In each loop circuit, the Hopf bifurcation gives rise to oscillations in activity (emerging limit cycle) whose frequency is a function of the axonal transmissions delays, membrane time constants, and synaptic time constants of the various synaptic connections of the loop (see "Materials and Methods, Theoretical analysis"). We compare the theoretically derived frequency of the oscillatory instability with the actual frequency of oscillations observed in simulations of each loop when it is isolated from the complete network in the rate model (see "Materials and Methods"). Synaptic weights of all connections in the loop circuit are set as to reach sufficient feedback to drive stable oscillations and measure the frequency of oscillations in the firing rate time course (see "Materials and Methods" for the definition of stable oscillation). The synaptic time scales of inhibition are widely debated, especially in the studies supporting the STN-GPe as the generator of β oscillations with values ranging from 7 to 20 ms (Plenz and Kital, 1999; Terman et al., 2002; Nevado-Holgado et al., 2010; Kumar et al., 2011; Pasillas-Lépine, 2013; Nevado-Holgado et al., 2014; Pavlides et al., 2015; Shouno et al., 2017; Koelman and Lowery, 2019; Chen et al., 2020). Therefore, we fix the decay time constant for the excitatory

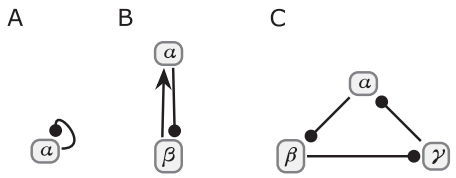


Figure 2. Theoretically investigated possible oscillation-generator networks. ***A***, A self-inhibiting population. ***B***, A reciprocal recurrent circuit comprising an excitatory and an inhibitory population. ***C***, Three populations create a loop with inhibitory connections.

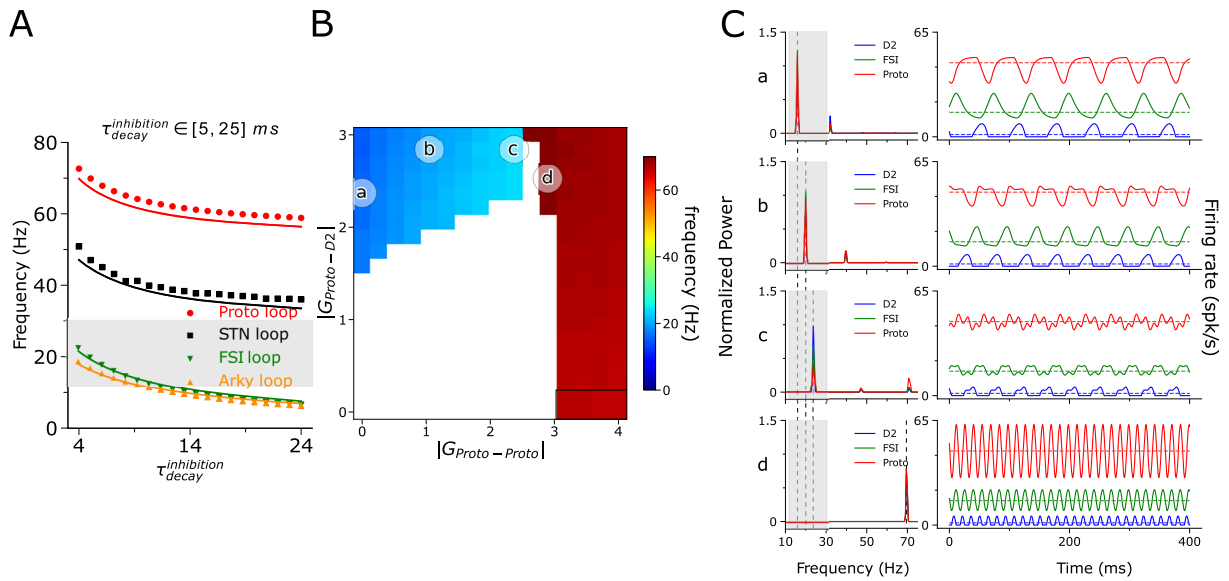


Figure 3. Pallidostriatal loop with the help of lateral Proto-Proto inhibition can generate β-band frequency oscillations in rate model. ***A***, Frequency of stable oscillations as a function of inhibitory synaptic decay time constant $\tau_{inhibition}$ for each individual loop. The shaded area indicates the β band (12–30 Hz). The scatter plot shows results from the rate model simulation whereas solid lines are estimates from theory (for more details, see "Theoretical analysis"). The range of values reported in experimental studies for the inhibitory synaptic time scale is shown above the plot. ***B***, Phase space of joint FSI and Proto loops. All the synaptic weights in the FSI loop increase along the *y*-axis though only the values of $G_{Proto-D2}$ are indicated ($G_{Proto-D2} = G_{D2-FSI} = G_{FSI-Proto}$). The color of each point in the phase space represents the frequency of stable oscillations corresponding to the network with those weights. Whereas, the white indicates that the network remains in steady state. Lastly, the hatched area indicates where *only* the Proto population has reached the stable oscillatory regime. ***C***, The normalized power spectrum (left) and mean firing activity of each population for the highlighted points in (*B*) (right). All data points represent a 20-s simulation of 100 neurons in each population with 0.1 ms time step.

projection from STN to Proto to $\tau_{\text{Proto-STN}} = 6$ ms and vary the decay time constant of inhibitory synapses τ_{inh} for all the inhibitory projections in the model in the range of 4–24 ms. For all values of τ_{inh}, we compare the theoretically derived frequency of the oscillatory instability with the frequency of oscillations observed in rate model simulations (Fig. 3*A*). The four considered loop circuits induce oscillations in a wide range of frequencies ranging from 5 to 80 Hz. Indeed, the oscillation frequencies derived from simulations are well predicted by our theoretical derivation. Importantly, across the wide range of inhibitory decay time constants considered here, the STN-GPe network does not produce β-band oscillations but rather oscillates in the γ range (35–50 Hz). Similarly, the short feedback provided by recurrent connections between Proto neurons drives γ-band oscillations with even higher frequencies (60–70 Hz). On the other hand, both pallidostriatal loops generate frequencies close to the lower boundary of the β-band (5–25 Hz; Fig. 3*A*).

Interactions between multiple BG circuits including the GPe but not the STN can lead to the emergence of β-like activity

In the full BG network, oscillations may be generated as a result of interactions between two or more loops. Since STN is not necessary for the induction of experimentally observed β-band oscillations in parkinsonian rats (De la Crompe et al., 2020), the pallidostriatal loops (FSI-D2-Proto loop and D2-Proto-Arky loop) may interact with the recurrent Proto loop to drive pathological oscillations in PD. Interestingly, the synaptic connections between FSI and D2 striatal neurons and the recurrent connections between Proto neurons are strongly affected by DD (Gittis et al., 2011; Miguelez et al., 2012), and see below for more details. Therefore, we investigate the possible interaction between the FSI-D2-Proto loop and the recurrent Proto loop in a reduced model comprising these two loops only. We vary the synaptic weights of the two loops $G_{\text{Proto-Proto}}$ and $G_{\text{Proto-D2}}$, run the simulation, and measure the frequency of stable oscillations in the firing activity (Fig. 3*B*). For low synaptic weights in either loop, the network does not leave the steady state. However, with increasing the G, the network is able to transition to the oscillatory regime (colored points in the phase space in Fig. 3*B*). Moreover, increasing the connection weight of the Proto loop shifts the frequency of the network from the lower bound of β to the higher end of this band (Fig. 3*C*). Altogether, our results show that the experimentally observed β-band frequency oscillations in the BG circuit could be resultant of the interactions between the pallidostriatal loops and a high-frequency oscillating di-synaptic loop.

Oscillation frequencies in a realistic network model of BG circuits with experimentally constrained time constants

The results of our rate model of BG circuits highlight the possible frequencies of oscillations driven by the various negative feedback loops of the network across a wide range of synaptic timescales and their possible interaction. However, neural dynamics in the rate model rely on a single time scale, ignoring the diverse timescales of neuronal and synaptic integration. The frequency of oscillations must thus be confirmed in a spiking neuronal network (SNN) where the membrane potential of the neurons is simulated using the LIF neurons (see "Materials and Methods" for details). In this more detailed model, all timescales—the synaptic rise and decay time constants, neuronal integration time constant, and axonal delays—are extracted from experimental observations (see Tables 3–5 for values and references to the corresponding experimental studies). Moreover, the SNN model will allow us to both produce realistic activity patterns and to directly compare the phase differences between the firing activity in the various neuronal populations during pathological β or artificially induced oscillatory activity (as in De la Crompe et al., 2020). The investigation of the normal and pathological dynamics of BG activity in the SNN will be guided by the general results obtained by analytical derivation and simulations in the reduced rate model. An example of the spiking activity in a single neuron from each population is demonstrated in Figure 4*A* (for presentation purposes, the APth is artificially aligned to 0 mV). After setting all the time constants and delays, we further verify the firing activity patterns of the network by mimicking the downstream effects of a short-term stimulation at the level of the motor cortex. Indeed, the temporal pattern of responses in the various neuronal populations of the BG is strongly coupled to the timescale of the synaptic and neuronal integration in the network. The effects of the cortical stimulation in all populations of our network thus offer an integrated view of the network's temporal properties that can be directly compared to experimentally observed responses (Kita and Kita, 2011). These effects include a direct excitation in both the STN

Table 5. Electrophysiological time constants for each synaptic connection

Projection	Δ (ms)	Reference	τ_{rise} , τ_{decay} (ms)	Reference
Proto-D2	6.89 ± 0.35 (4.3, 11.3)	Ketzef and Silberberg (2020) Kita and Kitai (1991)	0.80 ± 0.06, 6.13 ± 0.38	Sims et al. (2008)
STN-Proto	1.3 ± 0.3 (0.8, 2.5)	Kita et al. (1983)	1.1 ± 0.4, 7.8 ± 4.4	Baufreton et al. (2009)
Proto-STN	2.8 (2.0, 4.4)	Kita and Kitai (1991)	0.6 ± 0.1, 1.8 ± 2.5	Baufreton (unpublished)
Proto-Proto	4.67 ± 0.45 (3.05, 7.55)	Ketzef and Silberberg (2020) (mice)	0.50 ± 0.15, 4.91 ± 1.08	Sims et al. (2008)
Arky-Proto	4.55 ± 0.54 (2.55, 7.05)	Ketzef and Silberberg (2020) (mice)	0.50 ± 0.15, 4.91 ± 1.08	Sims et al. (2008)
D2-FSI	0.93 ± 0.29 (0.8, 2.0)	Gittis et al. (2010) (mice)	1.5 ± 2.9, 11.4 ± 2.1	Koos et al. (2004)
FSI-Proto	4.3 ± 0.7 (3.2, 7.0)	Glajch et al. (2016) (mice)	1.1 ± 0.4, 7.8 ± 4.4	Based on similarity to FSI-Proto
D2-Arky	4.9 ± 0.6 (3.8, 7.0)	Glajch et al. (2016) (mice)	1, 28	Baufreton (unpublished)

Axonal transmission delays, Δ, and the synaptic rise and decay time constants, τ_{rise} and τ_{decay}, are derived from the experimental literature referenced. Values are reported as mean ± SD. The range of Δ is also constrained by the data (same reference except for Proto-D2) as reported in the table.

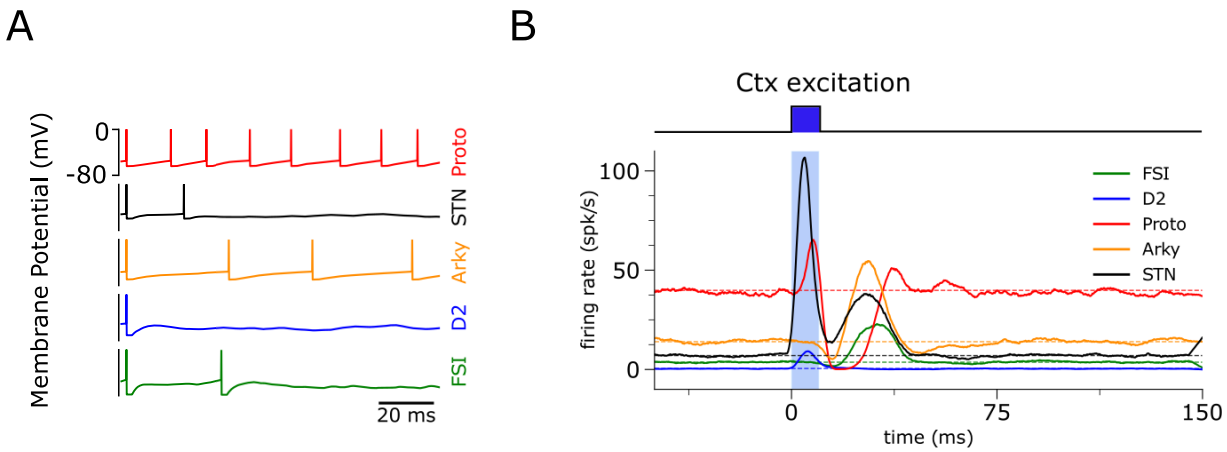


Figure 4. The SNN model captures electrophysiological characteristics of the BG network. ***A***, Example membrane potential traces of the LIF neurons in the model. The action potential threshold is artificially aligned to 0 mV for all neurons for presentation purposes. ***B***, The effect of 10 ms cortex stimulation on the population activity of the neurons in the model. Each time course is an average of 10 different stimulation protocols.

neurons as well as the D2-MSNs (Fig. 4*B*). Both of these inputs contribute to shaping the GPe activity. Kita and Kita (2011) describe the GPe response as, primarily, an early excitation (from STN), then, an inhibition (from D2-MSNs), and lastly, a late excitation (again from STN) followed by a late long-inhibition. Since Proto neurons make up the majority of the GPe population (Mallet et al., 2012), we attribute this prototypical response to the Proto population. Our model reproduces similar tri-phasic activity in the Proto population activity. In particular, the relative timing of the onsets and peak in activity in the various populations of the model agree with the observed pattern. There are, however, some slight differences. First, we do not observe the late and long-lasting inhibition in Proto observed by Kita and Kita (2011). This slow inhibition was shown to be driven mainly by GABA-b currents in Proto neurons, which are not included in our model. Since the time scale of GABA-b transmission is too slow (100–1,000 ms decay; Gerstner et al., 2014) to contribute significantly to the pathological β-band oscillations, we deliberately ignored this minor part of the inhibitory inputs to Proto. Second, the time scale of the re-stabilization of Proto activity after the last excitation is somewhat faster than what is observed experimentally. The slow recovery of Proto neurons may be explained by their specific intrinsic electrophysiological properties that may not be captured in the LIF model. These differences are unlikely to play a significant role in the generation or modulation of pathological β oscillations and are thus beyond the scope of this study.

Once all time constants and delays are set, we look at the frequency of the individual loops in the detailed SNN. We simulate each individual loop circuit when it is isolated from the network (Fig. 5*A*). In these simulations, the synaptic weights are set in a way to avoid the collective silencing of neurons in the population, while being sufficiently large to bring the circuit into the oscillatory regime (Fig. 5*B*). The temporal fluctuations in the mean activity across the neuronal populations reflect a clear oscillatory pattern while the high level of noise in the synaptic input induces large fluctuations within and between cycles, with waxing-and-waning oscillations in the mean activity. Importantly, the oscillations in mean activity are not reflected in regular spike trains firing at the oscillation frequency in single neurons (Fig. 5*C*). Rather, the probability of spiking for most neurons oscillates in time but the synaptic noise and network heterogeneities make it difficult (if not impossible) to detect oscillations at the level of individual neuron spike trains. Due to the low firing rate of D2 neurons, their contribution to oscillations is even more difficult to tease apart at the level of spike trains, even when the firing activities of many neurons are displayed simultaneously (Fig. 5*C*). Nevertheless, oscillatory activity can be clearly revealed by spectral analysis of the population activity. To quantify the resultant frequencies, we run multiple sets of simulations (with varying parameter initialization) for each network and then average their corresponding power spectrum densities (PSDs; Fig. 5*D*). The low-frequency component of the average population activity in our model may be reminiscent of local field potential recorded experimentally. The frequencies of oscillations generated in the SNN further confirm those of the rate model study. In particular, the recurrent connections between Proto neurons and the STN-GPe circuit generate γ-band frequencies (43 Hz for the STN-GPe circuit and 65 Hz for the Proto recurrent network). Whereas for realistic time constants, the two pallidostriatal loops oscillate with frequencies on the lower boundary of the β-band (Fig. 5*D*).

The asynchronous healthy state

In the previous section, we artificially segregated the various feedback loop circuits from the BG network and increased the strength of the synapses to bring each sub-circuit into an oscillatory regime. However, for each subnetwork, mild synaptic weight does not lead to the emergence of oscillatory activity. For the complete BG network, these mild synaptic connections lead to an asynchronous state where all populations display non-oscillatory fluctuations driven by the fluctuations in their synaptic inputs (synaptic noise) and the relative timing of spike occurrence across the neuronal populations (Fig. 6*A* and *B*). This non-oscillatory state can be mapped onto the healthy state of the BG network, with the mean and fluctuations of external inputs adjusted to drive the observed distributions of firing rate in the various BG populations of anesthetized rats (Mallet et al., 2005; Sharott et al., 2012, 2017; De la Crompe et al., 2020; Xiao et al., 2020). As the fluctuations in the network activity are non-oscillatory in this state, the CCs between the spike trains of pairs of neurons of a given population are flat in all populations of the network (Fig. 6*C*, see also Extended Data Fig. 6-1) and there is very low coherence between spike trains in a given

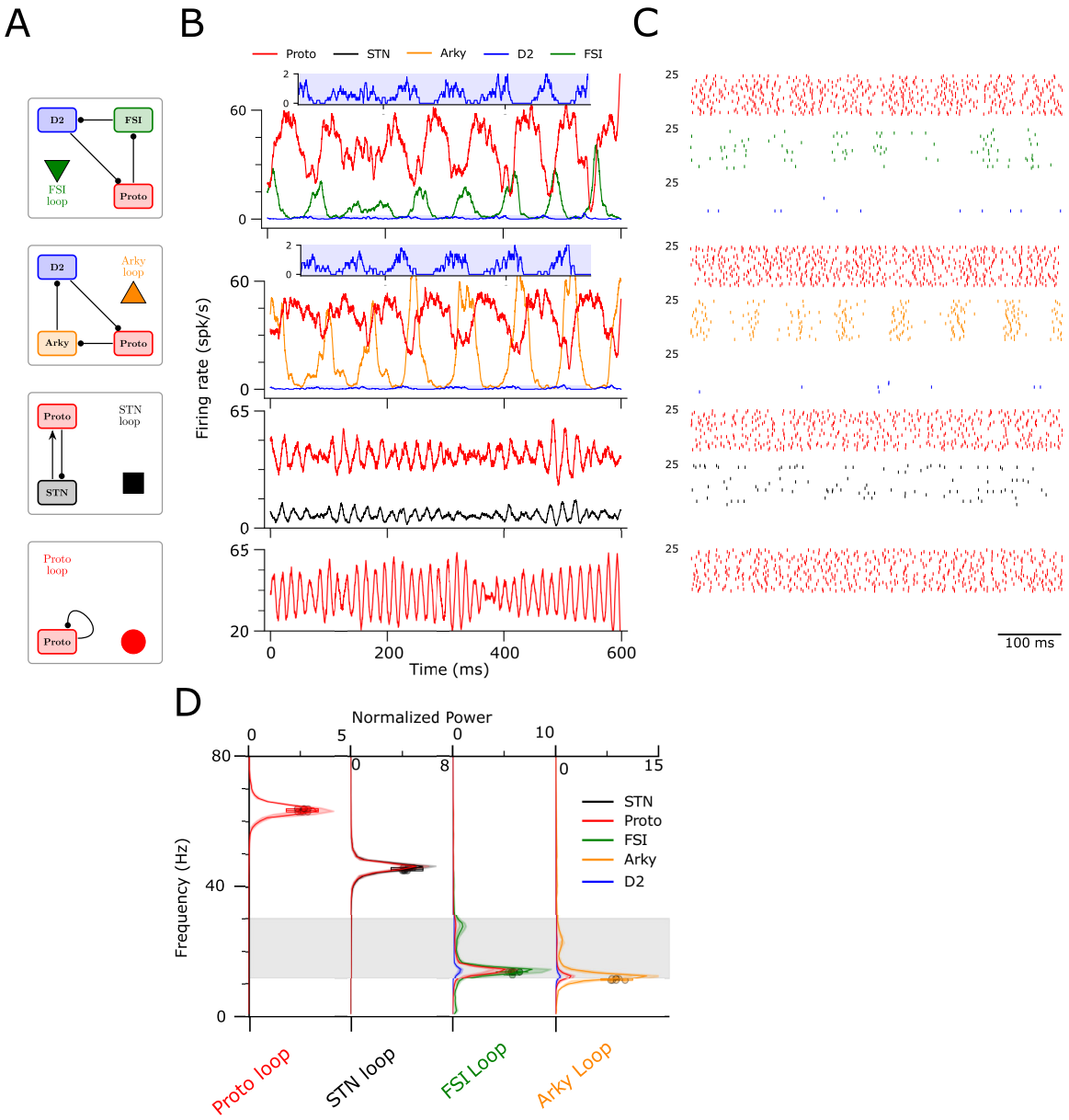


Figure 5. None of the individual feedback loops generate experimentally observed β-band frequency oscillations. ***A***, Schematics of individual feedback loops. ***B***, Simulations of individual feedback loops yield their intrinsic frequency. Example average firing rate of $N = 1{,}000$ neurons in each population of each given loop in (*A*). Insets highlight the activity of D2-MSNs where they exhibit oscillatory activity despite their low firing rate. ***C***, Individual spikes of $N = 25$ randomly chosen neurons in (*B*). ***D***, Normalized PSD of the mean population firing rate in each of the loops in (*A*) for 8, 10-s long simulations (see "Materials and Methods"). The box plots summarize the peak frequencies (represented by circles) measured for the Proto population. The shaded area indicates the β-band (12–30 Hz). Simulations are also repeated with 0.01 ms time steps (Extended Data Fig. 6-1).

population at any frequency (Fig. 6*D*, see also Extended Data Fig. 6-1) as has been experimentally revealed in the BG neuronal population of animals in healthy condition (Mallet et al., 2008b). While most parameters of the model are constrained by experimental data, the synaptic weights are mostly free parameters of the model until this point. In the first part of the Results section, we have shown that mid-β oscillations can be driven by the interaction between several oscillating circuits in the rate model (Fig. 3*B* and *C*). Multiple loop circuits must thus be at play to produce the experimentally observed β oscillations. Since the frequency of the resultant oscillations can change based on the contribution of each loop, it is now important to further constrain the synaptic weights in the model. While we cannot constrain all the synaptic weights with anatomical or physiological recordings, the weights along several connections of the circuits in the normal state have been set to reproduce the effect of artificial oscillatory drive by patterned optogenetic stimulation of the STN and Proto neurons in healthy rats reported in De la Crompe et al. (2020), as explained thereafter.

Activity modulation and phase differences during β-patterning

When β oscillations are artificially produced in healthy rats through the rhythmic activation of STN neurons with a 20 Hz half-sinusoidal optogenetic stimulation (Fig. 7*A*-left) (De la Crompe et al., 2020), Proto neurons display a fast excitatory response following the STN activation with a short delay. The amplitude of the modulations of STN and Proto activity in these

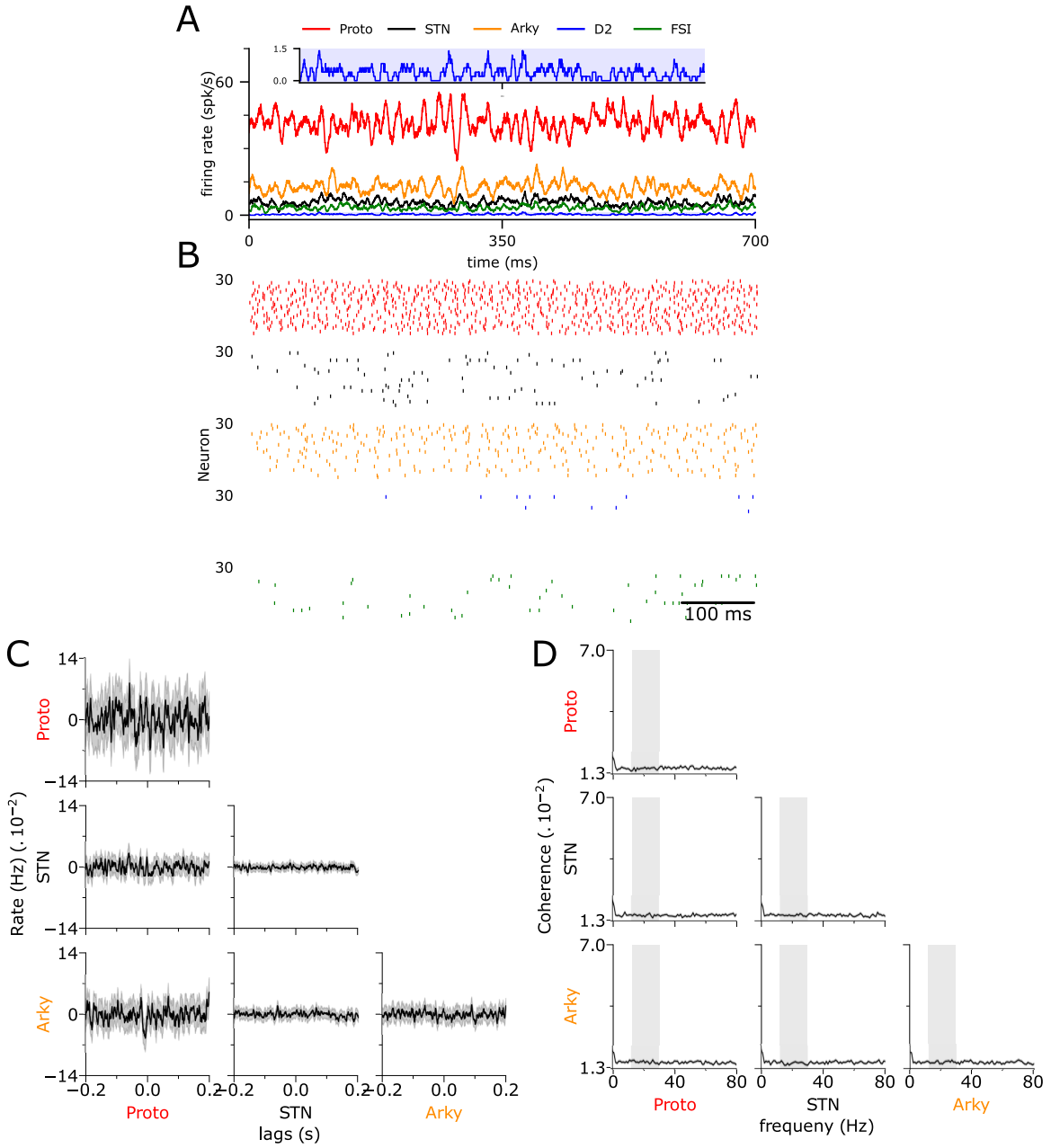


Figure 6. Healthy state of the tuned network does not exhibit any oscillatory activity. ***A***, Average population firing rate time course of neurons simulated in an anesthetized healthy state. $N = 1{,}000$ for all populations. Insets show the activity of D2-MSNs. ***B***, Individual spikes of $N = 30$ randomly chosen neurons in (*A*). ***C***, Average spike CC of 2,000 randomly chosen pairs of neurons from the two populations specified by the rows and columns. No significant CC is observed on average among the pairs of populations. ***D***, The average spike coherence of 1,000 randomly chosen pairs of neurons from the two populations specified by the rows and columns. The network simulation is run for 25 s and the coherence is measured using 1-s windows. The shaded gray area indicates the β range (12–30 Hz). No significant average spike coherence is observed on average among the pairs of populations. Coherence and CC of D2 and FSI populations are shown in Extended Data Figure 6-1.

experiments reflects the strength of the direct excitation from STN neurons to Proto. Moreover, during the same protocol, Arky neurons are inhibited by the Proto, with a modulation whose amplitude reflects the strength of the inhibition from Proto to Arky. We set the synaptic weights between STN, Proto, and Arky neurons to reproduce the respective modulation of their activity during this experiment by delivering a 20 Hz half-sinusoidal depolarizing input to the STN neurons (Fig. 7*A*-right). Note that the relative phase difference between the various populations during this patterned stimulation is also accurately reproduced due to the appropriate time constants of the model derived from other experimental data (see "Materials and Methods"). In another experiment, the β-patterning of Proto neurons' activity was induced in healthy rats through the non-specific rhythmic inhibition of GPe neurons with a 20 Hz half-sinusoidal optogenetic input that inhibits both the Proto and Arky populations (Fig. 7*B*-left) (De la Crompe et al., 2020). We reproduce this patterned stimulation in the model with a rhythmic inhibitory input onto Proto adjusted to yield the same level of minimum activity in Proto neurons as in the experimental data (Fig. 7*B*-right). The relative amplitude of Proto and STN modulation in this experiment constrains the STN-Proto synaptic

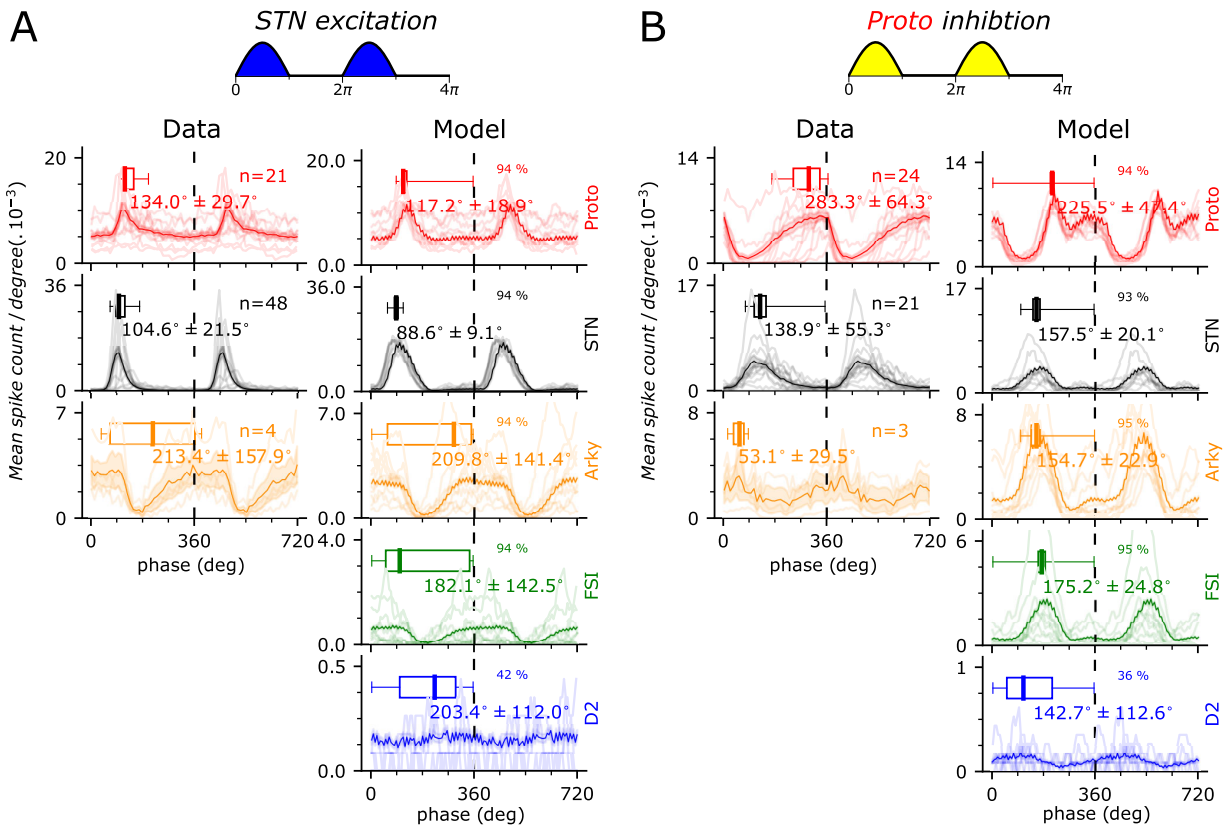


Figure 7. β-oscillation induction compared in the model and data. ***A***, β oscillatory activity is artificially induced by exciting STN neurons at 20 Hz frequency as indicated in the schematic on top. Left: Single neuron spike linear phase histogram of Proto (red), STN (black), and Arky (orange) neurons during artificial β-patterning using non-specific opto-inhibition of GPe neurons in healthy rats using eArchT3 (re-adapted from De la Crompe et al., 2020). Right: Single neuron spike linear phase histogram similar to (left), but for our tuned model including the FSI (green) and D2-MSNs (blue) while applying similar excitatory input as the optogenetic manipulation on the left. ***B***, Same as in (*A*) except for the fact that the β oscillatory activity is artificially induced by inhibiting Proto neurons at 20 Hz frequency as indicated in the schematic on top. In all panels, thinner lines represent randomly chosen individual neuron traces whereas solid line represents the average population phase histogram. The shaded area indicates the SEM. Box-and-whisker plots indicate median, first and third quartile, min and max values. In the model, only the neurons (out of the total of $N = 1{,}000$) that passed the Rayleigh test are taken into account; the percentage of which is indicated on the upper right quadrant of each panel.

Table 6. Population firing rates (Hz) during anesthesia

Neuron	FR (Ctrl)	Reference	FR (DD)	Reference
D2-MSN	0.49 ± 0.14	Sharott et al. (2017) Sharott et al. (2012)	2.80 ± 0.47	Sharott et al. (2017)
STN	7.0 ± 5.8	De la Crompe et al. (2020)	24.5 ± 11.8	De la Crompe et al. (2020)
Arky	14.1 ± 8.3	De la Crompe et al. (2020)	12.2 ± 6.0	De la Crompe et al. (2020)
Proto	39.8 ± 13.7	De la Crompe et al. (2020)	21.6 ± 12.2	De la Crompe et al. (2020)
FSI	3.67 ± 3.22	Mallet et al. (2005) Sharott et al. (2012)	4.93	Xiao et al. (2020)

Values are reported as mean ± SD. FR: firing rate; Ctrl corresponds to healthy animal and DD to dopamine-depleted state in 6OHDA-lesioned animals.

weight. Altogether, we are thus able to complete the tuning of the synaptic weights within the STN-GPe circuit and between GPe neuronal populations by replicating the β-patterning experiments. Note that as in Figure 4*B*, the stabilization of Proto activity following a transient excitatory input from STN neurons is slower in the experimental data compared to the model. Again, this is likely due to the intrinsic cellular properties of Proto neurons that are not adequately captured by our LIF model. Nevertheless, the Proto activity in the model is stabilized by the end of each stimulation cycle resulting in evoked responses in STN neurons fully compatible with the experimental data both in terms of phase relationship and amplitude of the response (Fig. 7*B*). Note that pallidostriatal and Proto-Proto weights are also tuned in this experiment however this tuning will be later explained in the next sections (Kita et al., 2004).

DD in the model gives rise to β oscillations

The depletion of dopamine leads to pathophysiological and anatomical changes in the BG (Mallet et al., 2019). These changes are reflected both in changes in firing rate distributions (see Table 6 for firing rates in the DD state) as well as some structural changes in the number and strength of synaptic connections in various BG pathways that are reported to occur after chronic DD. More precisely, (i) the number of connections in the projection from FSI to D2 neurons doubles (Gittis et al., 2011), (ii) the synaptic weights from Proto to STN neurons are doubled (Fan

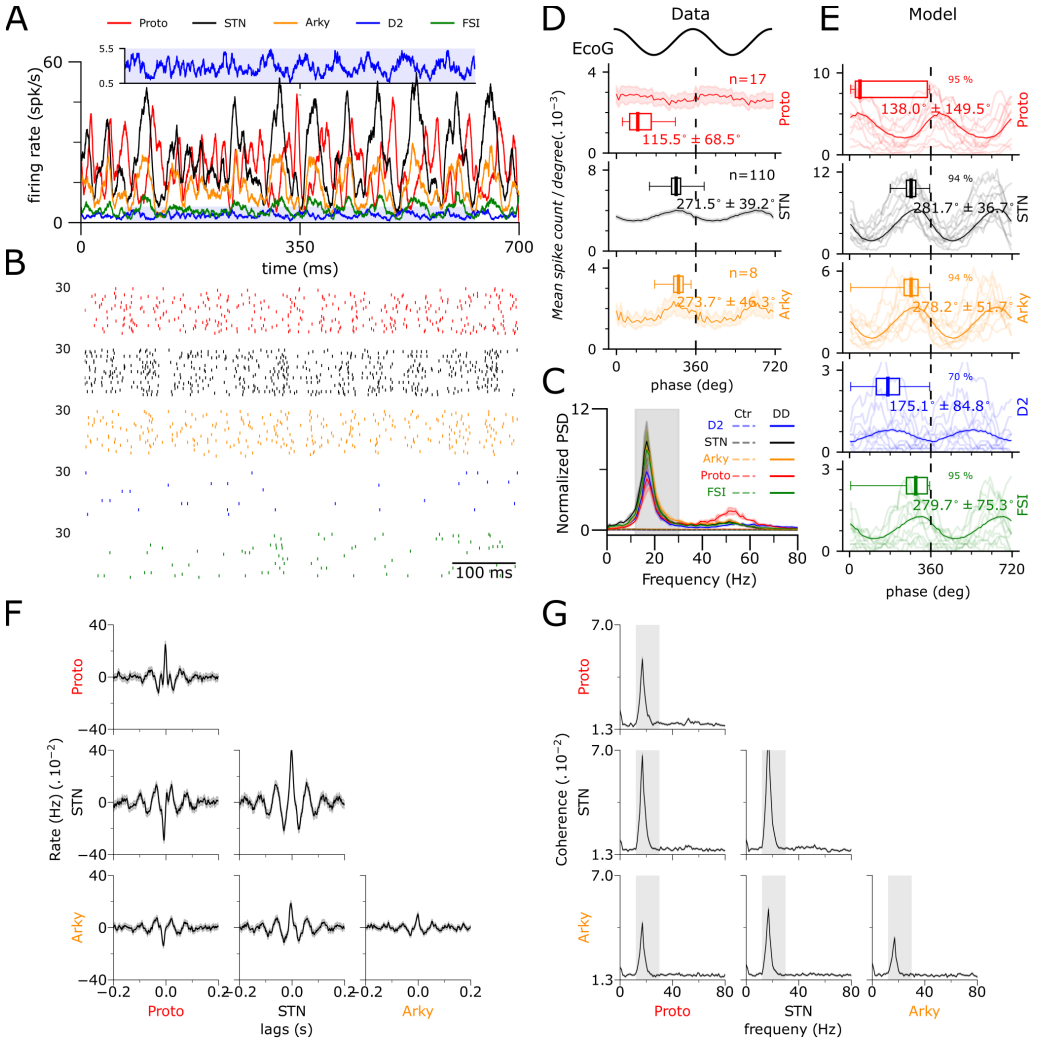


Figure 8. The model replicates the pathological phase locking and β-band oscillatory activity after transition to DD. ***A***, Average population firing rate time course of Proto (red), STN (black), Arky (orange), D2 (blue), and FSI (green) simulated in an anesthetized dopamine-depleted state. $N = 1{,}000$ for all populations. Insets highlight the activity of D2-MSNs where they exhibit oscillatory activity despite their low firing rate. ***B***, Individual spikes of $N = 30$ randomly selected neurons in (*A*). ***C***, Normalized power spectrum of population average firing rate in (*A*). A significant peak is indicated in the β-band (12–30 Hz, shaded in gray). ***D***, Single neuron spike linear phase histogram of Proto (red), STN (black), and Arky (orange) neurons during spontaneous β oscillations aligned to the recorded EcoG in anesthetized parkinsonian rats (re-adapted from De la Crompe et al., 2020). ***E***, Single neuron spike linear phase histogram similar to (*D*), but for our tuned model including the FSI (green) and D2-MSN (blue). Phases are aligned to Proto peaks after β filtering. Thinner lines represent results for randomly chosen individual neurons. The solid line represents the average population phase histogram and the shaded area indicates the SEM. Box-and-whisker plots indicate median, first and third quartile, min and max values. In the model, only the neurons (out of the total of $N = 1{,}000$) that passed the Rayleigh test are shown; the percentage of which is indicated on the upper right quadrant of each panel. ***F***, Average spike CC of 2,000 randomly chosen pairs of neurons from the two populations specified by the rows and columns. All population pairs exhibit significant spike CC with lags corresponding to the β-band frequencies. ***G***, The average spike coherence of 1,000 randomly chosen pairs of neurons from the two populations specified by the rows and columns. The network simulation is run for 25 s and the coherence is measured using 1-s windows. The shaded gray area indicates the β range (12–30 Hz). All population pairs exhibit enhanced average spike coherence in the β range. Coherence and CC of D2 and FSI populations are shown in Extended Data Figure 8-1. β oscillations result from synaptic changes due to DD in the FSI-D2-Proto loop that pushes it to the oscillatory regime (Extended Data Fig. 8-2). The effect of τ_m of STN neurons on frequency is also shown in Extended Data Figure 8-3. Network with the addition of Arky-D2/STN and FSI-Arky projections also exhibits β oscillations (Extended Data Fig. 8-4).

et al., 2012), and (iii) the synaptic weights of the recurrent inhibition between Proto neurons are doubled (Miguelez et al., 2012). We investigate the effect of DD in the BG by implementing all these functional and structural modifications in our BG network model. Following DD, the BG network sees the strength of the connections along several negative feedback loops strongly increased (doubled), namely the FSI-D2-Proto loop, the Proto-Proto loop, and the STN-Proto loop. The increased negative feedback along these loops is enough to push the network into an oscillatory regime. In this state, all the nuclei exhibit clear oscillatory activity that is observable in both the mean population firing rates as well as individual neurons' spiking patterns (Fig. 8*A* and *B*). To characterize the oscillatory activity, we run multiple 25-s-long simulations of the network initialized in the DD state. We then use each simulation to derive the PSDs of the mean population firing rates (for more detail refer to "Materials and Methods") and then average the PSDs over all repetitions. As shown in Figure 8*C*, the average PSD exhibits a significant increase in the β-band oscillatory activity in all populations after DD compared to the healthy state. The peak frequency of this oscillatory activity is 18 Hz. In addition, a small increase in the γ-band power (50–55 Hz) is observed in the Proto firing activity. The γ-band oscillatory activity in the Proto may give rise to the β–γ coupling observed in PD patients

and in rats (López-Azcárate et al., 2010; Dejean et al., 2011). We confirm the relative role of each sub-circuit in the generation of the observed oscillatory activity by simulating each sub-circuit individually. With the synaptic weights of the DD state, and while disconnected from the rest of the network, the FSI-D2-Proto loop displays strong oscillatory activity in the low-β frequency, while the Proto loop and the STN-Proto loop are both close to the oscillatory regime with low oscillatory content in the gamma range (Extended Data Fig. 8-2). Therefore, the pallidostriatal loop is the main driver of the observed pathological oscillations, and its interaction with either Proto-Proto or STN-Proto loop brings the oscillation in the mid-β range.

While the frequency of the oscillatory activity generated in our BG network in the DD state agrees with abnormal β-band oscillations, the nature of the interactions between the different populations that give rise to the β oscillations can be revealed by comparing relative phase distributions of spikes across all populations of the network. Indeed, De la Crompe et al. (2020) have reported the individual spike-phase histogram of the Proto, STN, and Arky neurons relative to the motor cortex electrocorticogram (ECoG) signal in parkinsonian rats (re-adapted in Fig. 8*D*). Their results provide a great biological reference for us to validate our model with. To compute the relative spike-phase histograms, we use the Proto mean population firing rate as our reference instead of the ECoG signal, as we do not include the cortex in our model. Using 25-s long simulations, we then calculate the relative phases of all the spikes for each individual neuron separately. In order to be able to compare the phase distribution to those of the experimental data, we make use of the findings of Sharott et al. (2017) where they show that in parkinsonian rats, the D2-MSNs exhibit anti-phase activity relative to the recorded ECoG. Therefore, we shift the phase distributions of all neurons so that the average phase distribution of D2-MSNs is anti-phase to the β-filtered ECoG signal in Figure 8*D*. The final results of the phase distributions for all the populations in the DD state are shown in Figure 8*E*. Our results show that the phase relationships between the Proto, STN, and Arky neurons in the model closely resemble those of the experimental data (Fig. 8*D* and *E*). In particular, peak STN activity is in phase with Proto inhibition, consistent with a main oscillatory drive coming from the GPe rather than from STN (see Fig. 7). The phase difference between Proto and STN argues against a significant contribution of the STN-Proto loop to the DD state oscillations observed here. To confirm the marginal (if any) role played by the STN-Proto loop, we also replicate the effect of manipulations of GPe and STN activity performed experimentally (see below). To our knowledge, the phase relationship of FSIs in the DD state has not yet been reported experimentally. As the FSI-D2-Proto sub-circuit plays a central role in the generation of the observed oscillatory activity, our model makes a strong prediction concerning the relative phase distribution of the spiking activity of FSI neurons relative to other BG populations that may be tested in future experiments.

In the DD state, the strong β-band oscillatory activity of the network can be revealed by CC analysis of the spiking activity of the Proto, STN, and Arky neurons (Fig. 8*F*). Indeed, the CC analysis shows a strong oscillatory pattern of correlation between neurons in a given population, as well as strong synchrony between the different populations. The peak and trough of the CCs at zero-lag also indicate the phase or anti-phase entrainment between the populations, confirming the previously presented phase difference analysis, with Arky and STN neurons oscillating in phase and Proto neurons displaying an anti-phase relation to each of them. This β-entrained correlation is evident in all the pairs of populations except for among the D2 neurons, due to their very low firing activity levels (Extended Data Fig. 8-1). The oscillatory synchronization between neurons within and across BG neuronal populations can also be revealed by the strongly increased coherence in the β-band between spike trains (Fig. 8*G*, Extended Data Fig. 8-1). GPe Arkypallidal neurons receive projections from D2-MSNs/STN neurons, as well as sending projections to FSIs (Fig. 1). However, these connections are all much weaker than corresponding projections to and from Proto (for STN-Arky vs. STN-Proto connections, see Ketzef and Silberberg, 2020; Pamukcu et al., 2020; for D2-MSN-Arky vs. D2-MSN-Proto, see Aristieta et al., 2021; for Arky-FSI vs. Proto-FSI, see Bevan, 1998; Saunders et al., 2016). Therefore, we simulate a network with the addition of those projections while estimating their number of connections as well as their average synaptic strengths at 20% of those measured for Proto. The results of these additional simulations show that the oscillation frequency and phase relationships of the activities remain unchanged (Extended Data Fig. 8-4). It is worth noting that increasing the weight of those connections results in abolishing the experimentally observed anti-phase locking of Arky and Proto, though their power spectrum would remain unaffected (data not shown here).

The abolishment of anti-phase coupling can be mentioned, but is not an argument against STN-Arky connections, since it could emerge from various factors not considered in this model, for example, thalamic input, different arrival times of striatal input at arkypallidal versus prototypical cells, or weak Arky-Proto feedback inhibition.

Proto inhibition, and not STN, abolishes β-band oscillations in the dopamine-depleted state

STN-GPe network has been a long-standing candidate for explaining the origins of the pathological β oscillations in the BG (Plenz and Kital, 1999; Terman et al., 2002; Nevado-Holgado et al., 2010; Kumar et al., 2011; Pasillas-Lépine, 2013; Nevado-Holgado et al., 2014; Pavlides et al., 2015; Shouno et al., 2017; Koelman and Lowery, 2019; Chen et al., 2020). However, De la Crompe et al. (2020) have recently shown that with collective opto-inhibition of the STN neurons, the β oscillations are still maintained at the level of both the cortex and GPe. In light of their results, it could be debated that the STN loop is not the generator of the β oscillations after all. In order to test this hypothesis, we inhibit the STN neurons using a uniform hyperpolarizing input that is applied to all the neurons in STN (Fig. 9*A*). This rather large inhibition does not lead to any significant change in the average firing rate of Proto neurons. This lack of modulation indicates that the synaptic feedback from STN to the Proto population is rather weak. We inhibit the network for 25 s and use the average population activity time courses of each population to measure their respective PSDs. Surprisingly, the STN inhibition not only does not diminish β oscillations in other populations, but it also boosts the oscillation power in the β-band (Fig. 9*B*). On the other hand, understandably, the reduction in STN activity due to the inhibition results in a decrease in the power of its β-band oscillations (Fig. 9*B*).

In a similar experiment as above, De la Crompe et al. (2020) targeted the whole GPe population using a non-specific inhibitory opsin. The Opto-inhibition of the GPe neurons abolished the β-oscillation in activities of both the motor cortex ECoG and STN activity. Their results suggest a key central role for

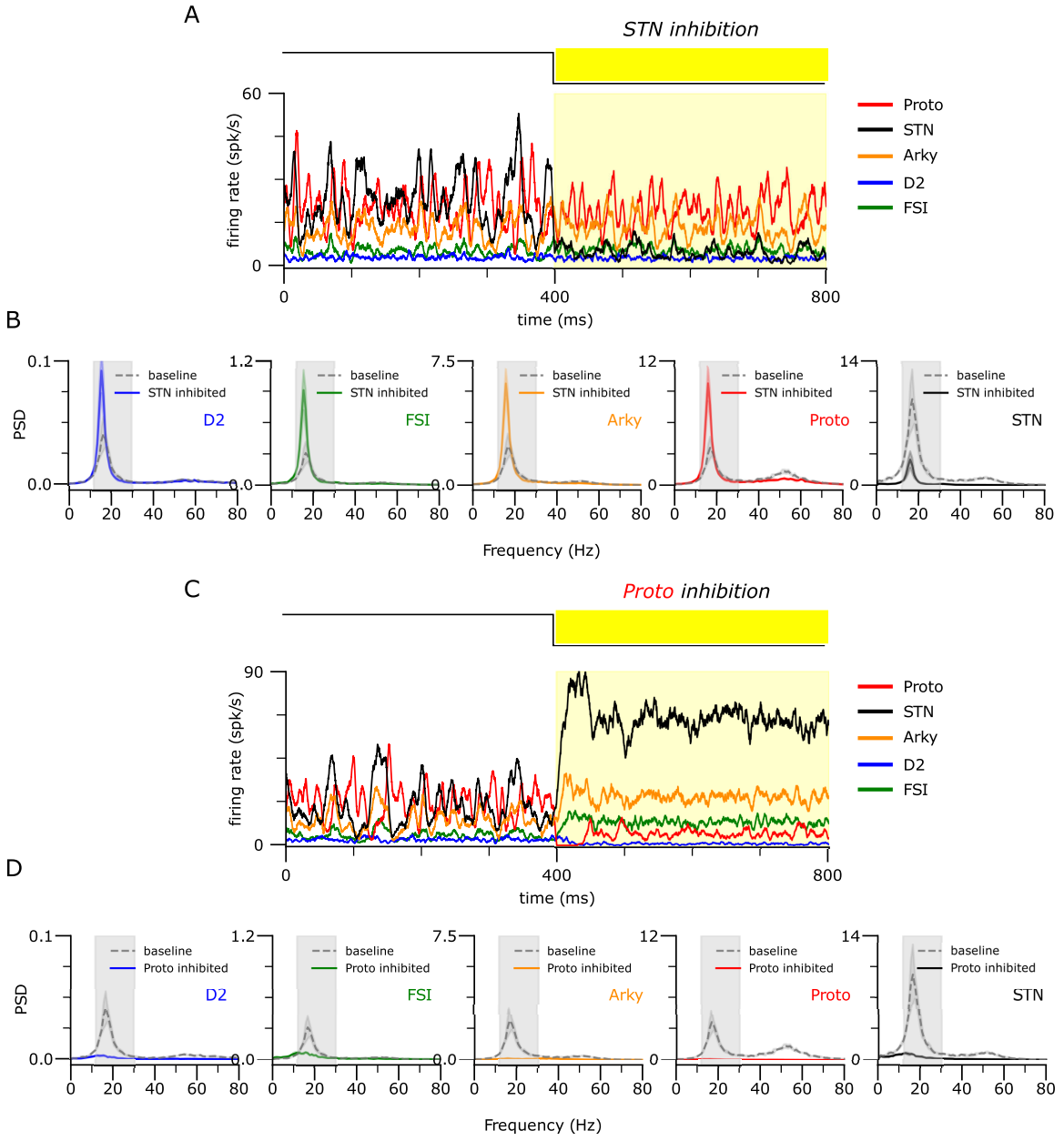


Figure 9. Proto inhibition, and not STN, abolishes β-band oscillations in dopamine-depleted state. ***A***, Average population firing rates in a network of Proto, STN, Arky, D2, and FSI when in a dopamine-depleted state (DD) the STN neurons are inhibited. The yellow shaded color indicates where external inhibitory input is applied to STN. ***B***, PSDs of each of the populations compared between baseline in DD (dashed gray) and when inhibition is applied to STN (solid colored). STN inhibition does not uplift β-oscillation. ***C***, Average population firing rates in a network of Proto, STN, Arky, D2, and FSI when in a dopamine-depleted state (DD) the Proto neurons are inhibited. The yellow shaded color indicates where external inhibitory input is applied to STN. ***B***, PSDs of each of the populations compared between baseline in DD (dashed gray) and when inhibition is applied to Proto (solid colored). Contrary to STN, Proto inhibition abolishes β-oscillation in all populations. Gray shaded area indicates the β range (12–30 Hz).

the GPe in the generation of the β oscillations. Therefore, we hypothesize that since the Proto population is the shared node in all our loops, its inhibition should in principle abolish all oscillatory activity in the DD state. We then tested this hypothesis by inhibiting all the Proto neurons using a uniform hyperpolarizing input. Contrary to STN, Proto inhibition significantly affects the STN activity via disinhibition. This is partly due to the increase in STN-Proto weight after DD (Fan et al., 2012; Fig. 9*C*). The power spectrum density analysis shows that the Proto inhibition results in total suppression of β-oscillation in the activity of all the populations as shown in Figure 9*D*. These findings rule out the significant contribution of the STN loop in the generation of β oscillations. In fact, they support the hypothesis that the Proto–central loops are orchestrating the β oscillations in the DD state.

Discussion

Building on up-to-date experimental evidence, our model shows that the neuronal and synaptic properties of dopamine-depleted pallidostriatal circuits can fully explain the β synchrony observed in experimental parkinsonism. The loops linking pallidal (Arky and Proto) and striatal (FSI and D2) neuronal populations generate oscillations in the low-β range (<15 Hz). However, the

interactions between these loops and strong inhibitory reciprocal connections within the pallidal Proto neurons can increase the oscillatory frequency in the complete network, driving abnormal β oscillations with experimentally observed frequencies (~20 Hz). Consistent with data, inhibition of the GPe, but not the STN, impairs the expression of abnormal β oscillations (De la Crompe et al., 2020). Indeed, the two GPe neuronal populations play a crucial role in the generation of pathological oscillatory activity in our model. These neurons participate in the two pallidostriatal reciprocal and recurrent circuits which induce β oscillations after DD. The GPe having a central role in the circuitry also imposes phase relationships between various BG populations that are consistent with those reported in parkinsonian animals and when β oscillations are artificially driven through GPe activity modulation (De la Crompe et al., 2020). On the contrary, the timescale of activity propagation in the STN-GPe network is inconsistent with a putative generation of β rhythms and the observed strength of its reciprocal connections is too low to be the main drive of oscillatory activity (Extended Data Fig. 8-2).

The large amount of data collected in parkinsonian rats, including the phase relationships between BG neuronal populations during β oscillations and the effects of neuronal manipulation on abnormal β, constrains the underlying mechanisms as shown here. We thus chose to build our model according to the data collected from healthy and parkinsonian rats (and, when unavailable, from mice). Future experiments uncovering phase relationships between BG neuronal populations during abnormal β oscillations in patients or non-human primates (NHPs) could inform us about a possible parallel in the β-generation mechanisms between parkinsonian patients, NHPs, and rats. Parkinsonian rats display abnormal β-activity that is highly behavior and state-dependent (Mallet et al., 2008a; Avila et al., 2010; Brazhnik et al., 2021) and can spread from mid-β (15–25 Hz) under anesthesia (Mallet et al., 2008a; De la Crompe et al., 2020) to high-β (25–35 Hz) in awake behaving parkinsonian rats (Bergman et al., 1990; Avila et al., 2010; Brazhnik et al., 2014; Delaville et al., 2015; Brazhnik et al., 2016). The interaction between multiple oscillation generation circuits, as proposed here, naturally accounts for frequency changes as the relative strength of each oscillatory circuit can influence the frequency of global oscillations within the β range (Fig. 3*C*).

The mechanism for abnormal β oscillations revealed here may also be at play in parkinsonian patients and in other animal models. However, the properties of abnormal β activity differ between animal models and different generation mechanisms may be involved. First, the frequency of abnormal β-oscillations under DD varies across species (Stein and Bar-Gad, 2013; Smith and Galvan, 2018). Parkinsonian mice mostly show increased field potential activity in the delta range when active and are not reported to exhibit exaggerated β activity (Willard et al., 2019; Whalen et al., 2020) (but see Baaske et al., 2020). NHP models of PD display excessive β with peak frequencies in a reduced low-β range (Nini et al., 1995; Raz et al., 2001; Stein and Bar-Gad, 2013; Deffains and Bergman, 2019). In PD patients, the frequency of abnormal β activity varies across individuals (Kühn et al., 2009), with local field potential recordings typically showing exaggerated β-range activity with frequencies within the low (10–15 Hz) and/or high (20–30 Hz) bands (Hutchison et al., 2004), possibly reflecting different generation mechanisms. Second, previous experiments suggest that STN plays a central role in abnormal β generation in NHP (Tachibana et al., 2011; Deffains and Bergman, 2019) while it does not in parkinsonian rats (De la Crompe et al., 2020). A circuit involving the STN, such as STN-GPe loop or the hyperdirect loop (Leblois et al., 2006; Pavlides et al., 2015), may thus contribute to the low-β observed in NHP. A recent modeling study suggests that STN-GPe and pallidostriatal circuits could interact to drive such abnormal β oscillations (Ortone et al., 2023). The multiple bands of β activity observed in PD patients may reflect a combination of mechanisms only partially reflected in the various animal models. Alternatively, differences in the relative strength of the pallidostriatal and Proto loops leading to oscillation frequencies (see Fig. 3*C*) may account for different frequencies between patients.

While pallidostriatal circuits can generate pathological β oscillations showing characteristics of those observed in parkinsonian rats, we cannot exclude other circuits taking part in β generation. In particular, the large BG-thalamo-cortical loop circuits (Leblois et al., 2006; Pavlides et al., 2015) or the STN-GPe loop may interact positively with BG-generated β oscillations (Corbit et al., 2016; West et al., 2022; Ortone et al., 2023). However, experiments in parkinsonian rats clearly show that the cortex and STN are not necessary for the expression of β oscillations while in fact GPe is (De la Crompe et al., 2020). It leaves a possibility for BG-thalamic loops (Str-GPe-SNr-Thal-Str) to interact with the presently investigated circuits (Brazhnik et al., 2016). GPe arkypallidal neurons project to striatal FSIs, an anatomical connection not considered in our main model. It could oppose the negative feedback in the Arky-MSN-Proto loop. However, this connection is considerably weaker than the projections of Proto onto FSI (Corbit et al., 2016) and is thus unlikely to impede or significantly alter oscillation generation (Extended Data Fig. 8-4). Within the pallidostriatal circuit, interactions between Proto and Arky may also participate in fast oscillations generated by the GPe alone (Gast et al., 2021).

Many previous models have been proposed to explain the generation of beta oscillations in the BG, with origins of the oscillations in the STN-GPe loop (for review, see Pavlides et al., 2015; Rubin, 2017). Given the initial observation of spontaneous low frequency (2 Hz) oscillations involving STN rebound following offset of GPe inhibition in STN-GPe organotypic cultures (Plenz and Kital, 1999), Terman et al. (2002) proposed a model network of conductance-based STN and GPe neurons, tuned based on in vitro rat data, where a similar mechanism could generate tremor-frequency (3–8 Hz) oscillations. Many following modeling studies proposed that the STN-GPe loop can generate beta-band oscillations (up to 30 Hz; Nevado-Holgado et al., 2010; Kumar et al., 2011; Park et al., 2011; Merrison-Hort et al., 2013; Pasillas-Lépine, 2013; Pavlides et al., 2015; Shouno et al., 2017; Koelman and Lowery, 2019). Some models rely on long synaptic transmission delays or time constants (Nevado-Holgado et al., 2010; Pasillas-Lépine, 2013; Pavlides et al., 2015; Shouno et al., 2017; Koelman and Lowery, 2019) or slow neuronal integration time constants (Kumar et al., 2011). We believe that our careful estimation of the timescale of synaptic transmission and neuronal integration in the BG network makes these putative generation mechanisms unlikely. Other models (Park et al., 2011; Merrison-Hort et al., 2013) rely on multiple interacting ion channels whose role in beta remains to be confirmed experimentally. Given the known properties of the STN-GPe circuit in rodents, we believe that it is unlikely to drive beta oscillations by itself in parkinsonian rats. Alternatively, previous models have proposed beta generation mechanisms in the cortex (Brittain and Brown, 2014) and along the hyperdirect pathway (Leblois et al., 2006). While experiments performed in rats seem to rule out a significant contribution of the cortex and STN in abnormal

beta oscillations (De la Crompe et al., 2020), these mechanisms are unlikely to play a significant role in parkinsonian rats. Finally, network mechanisms in the striatum (McCarthy et al., 2011; Kondabolu et al., 2016) may also be involved in beta generation. The adequation between the timescales in the striatal network and the observed beta may be further analyzed in future modeling work to confirm a possible contribution of the striatum.

Our model makes clear predictions concerning the phase of BG populations during β oscillations that can be tested both in rodents and in NHP. On one hand, the phase of FSI firing, yet unknown in rodents and a critical player in our model, may motivate future experiments in parkinsonian rats. On the other hand, the phase relationships between STN neurons and the two GPe populations recently revealed in NHPs (Katabi et al., 2022) can be quantified in future studies to test whether similar mechanisms are at play in this animal model. Synchronization at 10–13 Hz emerges in the BG-thalamo-cortical network of genetic absence epilepsy rats from Strasbourg (GAERS) during spike-and-wave discharges (SWDs) characteristic of absence epilepsy seizures (Paz et al., 2005). The phase relationship between STN and GPe neurons in SWD is similar to their phase relationship during parkinsonian β oscillations (Paz et al., 2005; De la Crompe et al., 2020). The pallidostriatal loop, which can produce low-β rhythm as shown here, may also be involved in such pathological oscillatory activity, possibly amplified or complemented by the larger hyper-direct loop (Arakaki et al., 2016). DD induces a number of modifications in the BG network, some of which are likely causal to the emergence of abnormal oscillatory activity (Quiroga-Varela et al., 2013).

Our model incorporates functional (e.g., changes in excitability) and structural (remodeling of synaptic connectivity) changes reported in the pallidostriatal and pallidosubthalamic circuits. In particular, the number of synapses from FSI to D2 (Gittis et al., 2011) as well as the strengths of the synapses in STN-GPe and GPe-GPe connection are increased (Fan et al., 2012; Miguelez et al., 2012). Unlike immediate functional changes induced by the lack of dopamine (e.g., changes in D2 excitability or the modulation of cortico-striatal synaptic transmission), these structural changes rather appear over days and are regarded as homeostatic processes that compensate for immediate activity changes induced by dopamine loss (Gittis et al., 2011; Ketzef et al., 2017; Villalba and Smith, 2018). Correspondingly, the observed hyperactivity of D2-MSNs after DD and predicted by the classical model of PD (Albin, 1989) are reversed over days following DD (Parker et al., 2018), possibly due to the increased inhibition from FSI after synaptic remodeling. The emergence of pathological oscillatory activity is induced in our model by the structural changes in the D2-FSI connections and is thus expected to arise with a similar delay following immediate DD. This is consistent with the relatively late emergence of pathological β oscillations in these models (Leblois et al., 2007; Degos et al., 2009; Quiroga-Varela et al., 2013).

Our modeling study provides a candidate mechanism for the generation of β abnormal activity in PD, compatible with evidence in parkinsonian rats, and makes clear predictions to be tested both in rats and other animal PD models. The variety of properties of β oscillations observed in PD patients and animal models (frequency band and phase relationships) may reflect very different generation mechanisms. As the link between abnormal synchronizations of neuronal activity and abnormal movement in PD is still controversial (Kühn et al., 2006; Leblois et al., 2007; Quiroga-Varela et al., 2013; Sharott et al., 2014; Neumann et al., 2016), it is crucial to uncover the neural mechanisms for the generation of oscillatory activity in each condition and context to improve our understanding of the pathophysiological mechanisms underlying PD and develop new therapeutic strategies.

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